assignments in

Junior Science

book 5

Waves

CHRIS GREEF
SUADA BILALI

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assignments in

Junior Science

book 5

Waves

CHRIS GREEF
SUADA BILALI

INSTANT LESSONS

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Introduction

Each book in the *Assignments in Junior Science* series provides a selection of worksheets covering one or two major topics. The worksheets are independent, supplying most information necessary to formulate answers. In the answer section at the back of each book, sample answers to most tasks have been given.

As students work through these activities, they will interpret and analyse data including text, tables, graphs and diagrams; and organise and represent data in a variety of formats. They will also utilise basic scientific skills such as working with formulas, completing calculations, making generalisations and explaining what they have discovered. The activities incorporate a range of theoretical and practical scientific skills, and also involve the development of literacy skills.

Suada Bilali and Chris Greef are both experienced and practising science teachers who had a lot of fun putting together this collection of worksheets.

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Images in a concave and in a convex mirror 1a

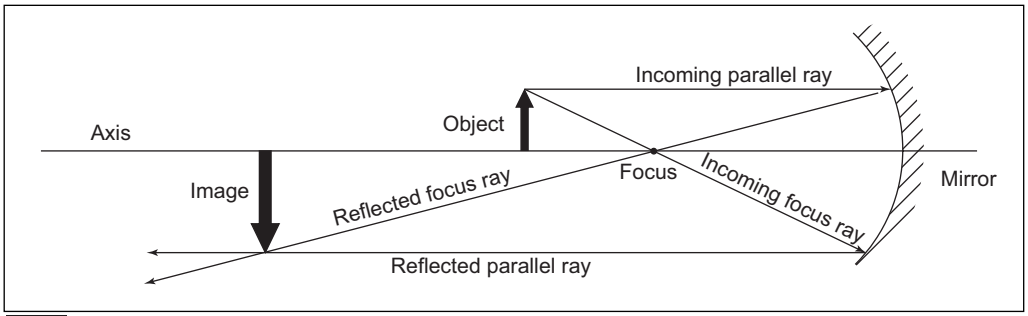
It is possible to predict exactly where the image of an object will appear when that object is in front of a curved mirror. If you select a point on an object (traditionally the tip of a vertical arrow), draw a couple of light rays from that point and follow some rules about reflection, you can find the position of the reflected arrow tip. The position where the two light rays from the object meet or appear to meet after reflection fixes the image.

Here are some things about light rays that you need to know to draw concave mirror diagrams:

- A parallel ray is a ray that is parallel to the axis.
- A focus ray is a ray that travels through the focus.
- An incoming ray travels from the object to the mirror.
- A reflected ray travels away from the mirror.
- After an incoming parallel ray is reflected, it becomes a reflected focus ray.
- After an incoming focus ray is reflected, it becomes a reflected parallel ray.
- If an incoming ray does not hit the mirror, extend it backwards.
- Reflected rays that don't seem to meet might do so if they are extended backwards.
- If reflected rays do not meet, a clear image is not produced.
- If the image is on the same side of the mirror as the object, it is a real image.
- If the image is on the opposite side of the mirror to the object, it is a virtual image.

Compared to its object, an image can be closer to the mirror, further away or the same distance from the mirror. It can be smaller, larger or the same size. It can be upright or inverted, and real or virtual.

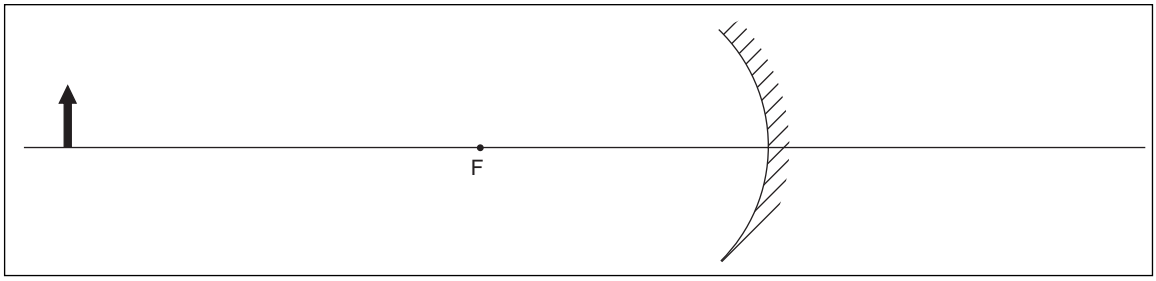
The following diagram shows how drawing a parallel ray and a focus ray from the tip of an arrow can be used to position the reflected tip of the arrow:



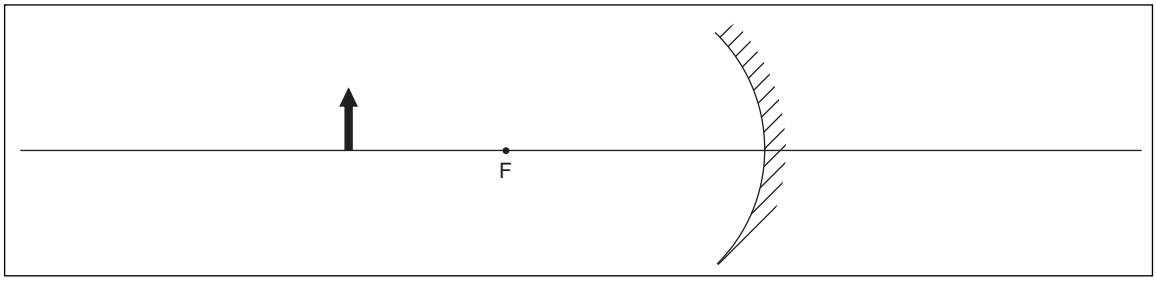
The image is further from the mirror than the object, larger than the object, inverted and real.

1 Construct the image of the object shown in each diagram. Compare image and object.

a



b



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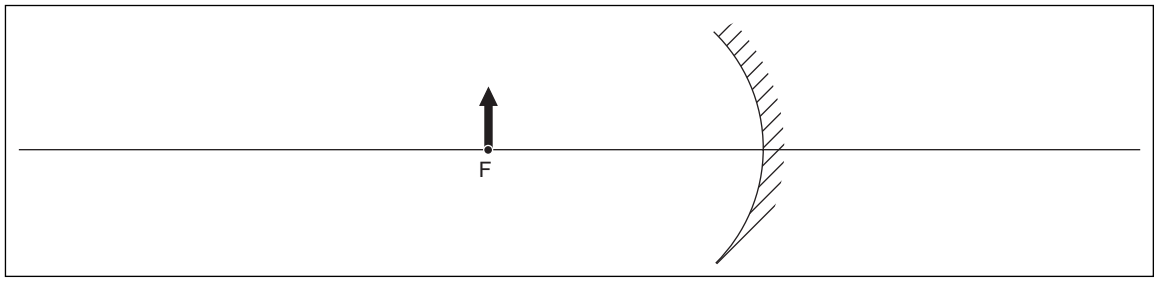
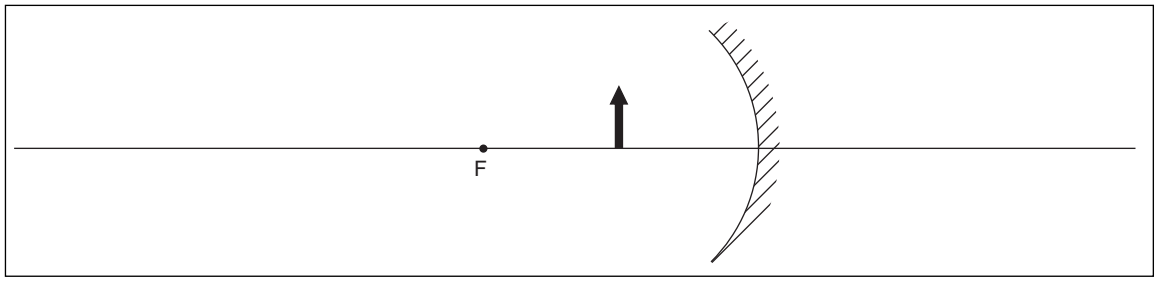
Images in a concave and in a convex mirror

1b

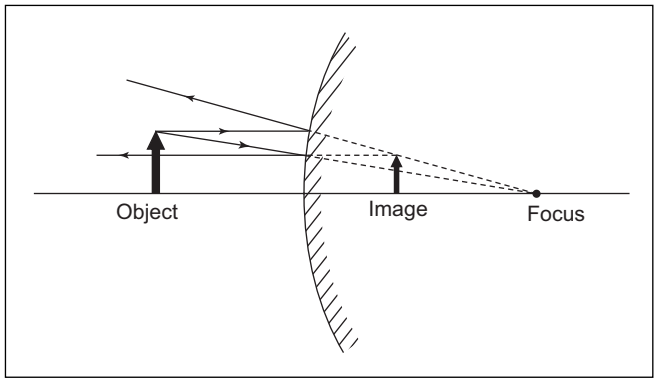
c



d



Below is an object–image diagram for a convex mirror.



The image is closer to the mirror, smaller than the object, upright and virtual.

2 **a** Develop a set of reflection rules to be followed when drawing convex mirrors.

b Draw another object–image diagram for a convex mirror in the box at right. Have the object in a different position to that in the diagram above. Apply your rules. Describe the image.

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Objects and their images in a lens

2a

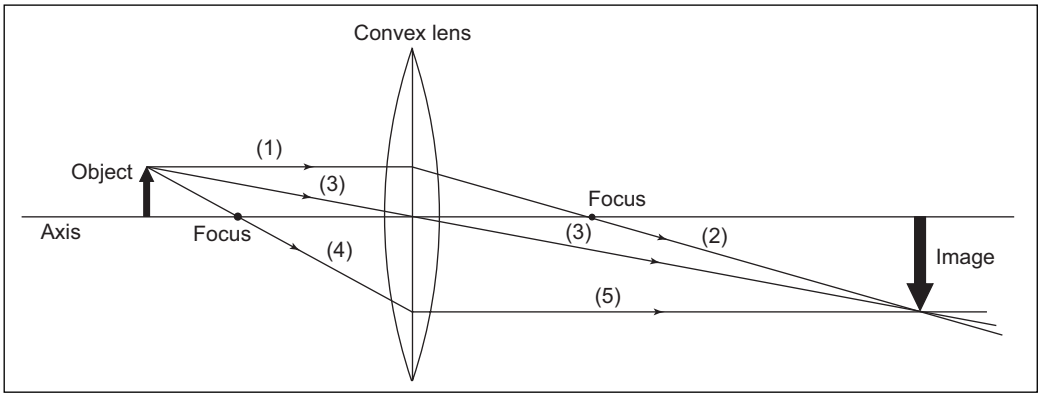
If you want to see an object clearly when using a lens, you have to hold the lens a certain distance away from the object and a certain distance from your eye. It is possible to predict where the focused image will be, if the object is a known distance from the lens. Of course when using a lens, you just move your eye to where the image is clearest and sharpest. To position images by drawing, a couple of rays from a point on an object (e.g. the tip of an arrow) are selected, and drawn following some rules about refraction. The refracted image position of the arrow tip will be given by where these rays meet.

Here are some things about light rays that you need to know to draw images in a convex lens:

- A parallel ray is a ray that is parallel to the axis.
- A focus ray is a ray that travels through the focus.
- A centre ray is a ray that passes through the centre of the lens.
- An incoming ray travels from the object to the lens.
- A refracted ray travels away from the lens.
- Light rays should be drawn as though they refract (i.e. bend) at the middle of the lens.
- After an incoming parallel ray (1 in the diagram below) is refracted, it becomes a refracted focus ray (2).
- An incoming centre ray (3) continues in a straight line through the lens.
- After an incoming focus ray (4) is refracted, it becomes a refracted parallel ray (5).
- If incoming rays do not reach the lens, extend them backwards.
- If refracted rays do not seem to meet, extend them backwards.
- If refracted rays do not meet, a clear image is not produced.
- If the image is on the other side of the lens to the object, it is a real image.
- If the image is on the same side of the lens as the object, it is a virtual image.

Compared to the object, an image can be closer to the lens, further from or the same distance from the lens as the object; smaller than the object, larger or the same size; upright or inverted; and real or virtual.

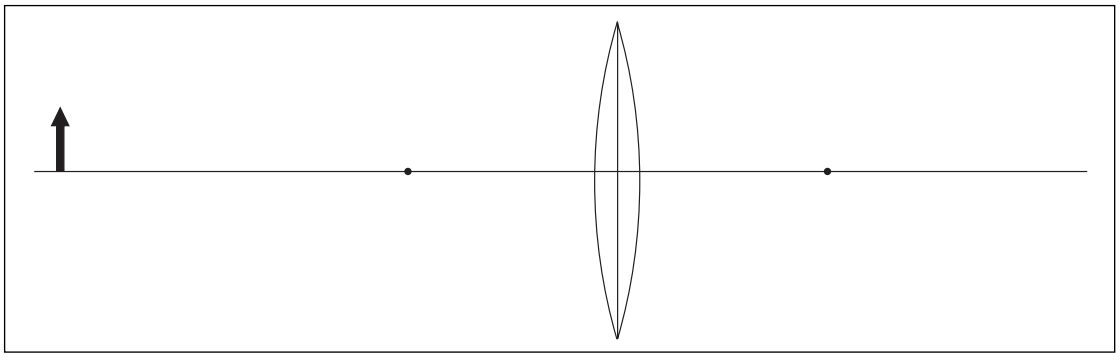
The diagram shows how the image for an object placed near a convex lens can be drawn.



The image is further from the lens, larger than the object, inverted and real.

1 Construct the image of the object in each of the following diagrams. Follow the rules listed above. Compare the image with its object.

a



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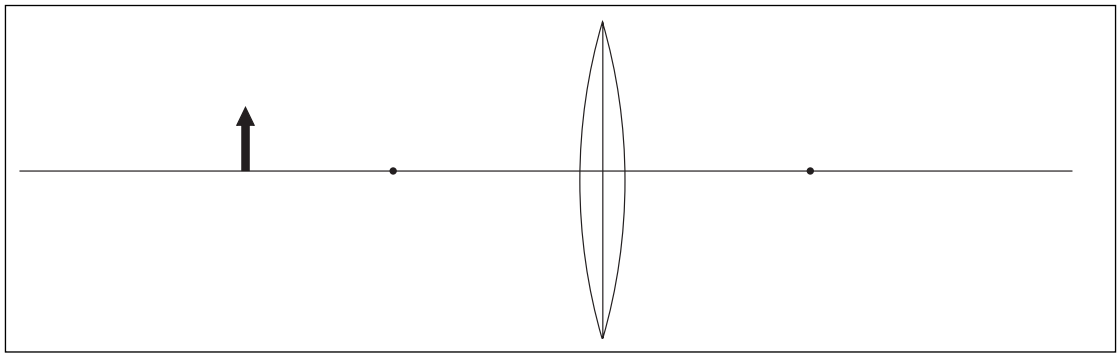
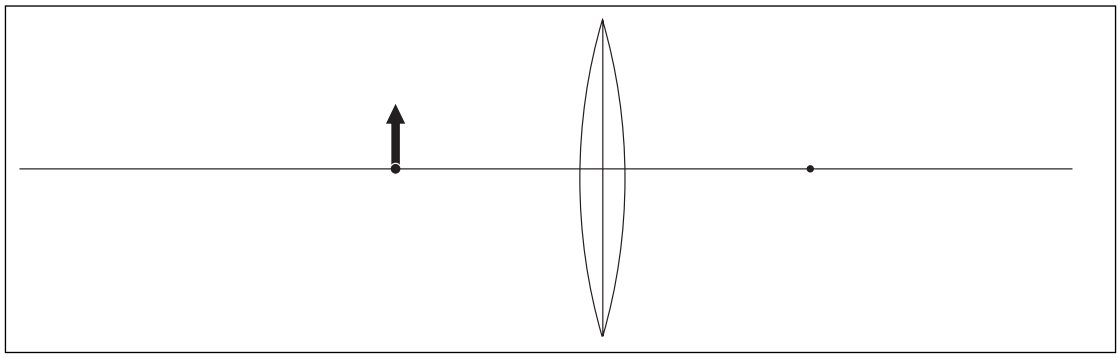
Objects and their images in a lens

2b

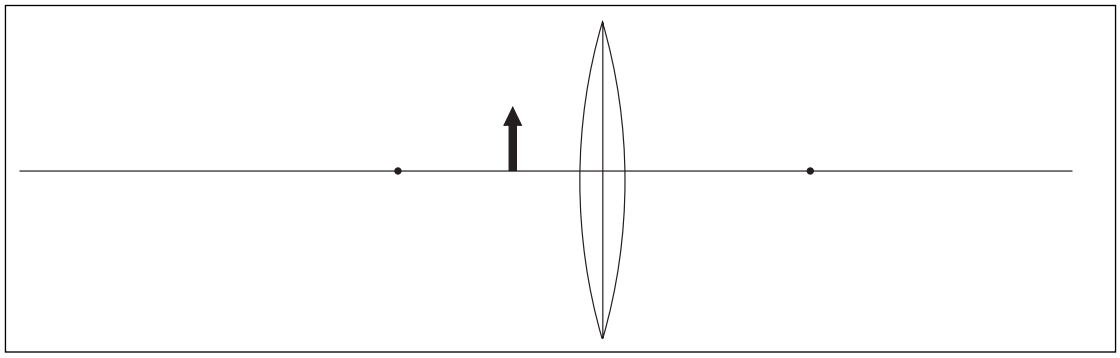
b



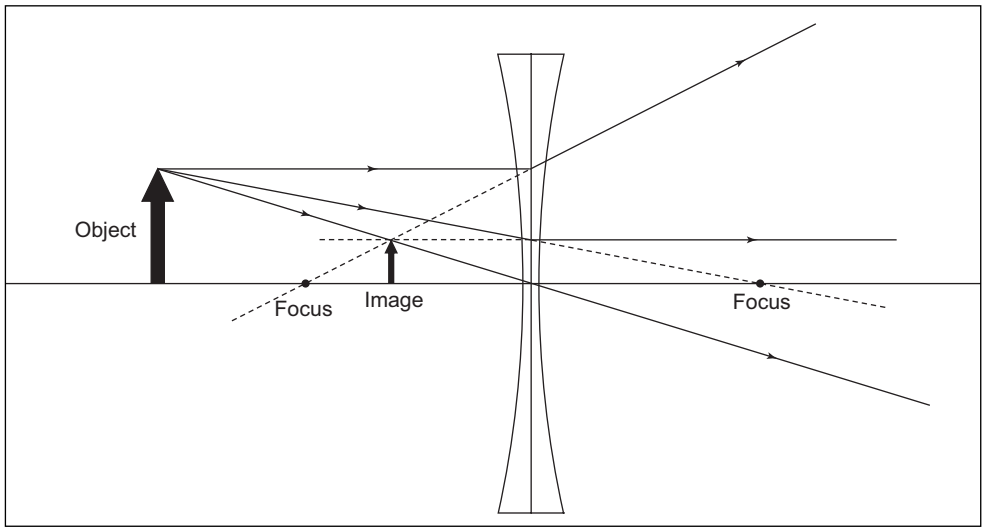
c



d



Below is an object–image diagram for a concave lens. The diagram shows the paths of three significant rays that are important in the drawing of the image.



The image is closer to the lens than is the object, smaller than the object, upright and virtual.

2 Study the diagram and develop a set of refraction rules that will help you draw images produced by concave lenses. Then construct a similar diagram with the object in a different position and apply your rules to find and describe the image. Do these tasks in your science book.

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The electromagnetic spectrum

3a

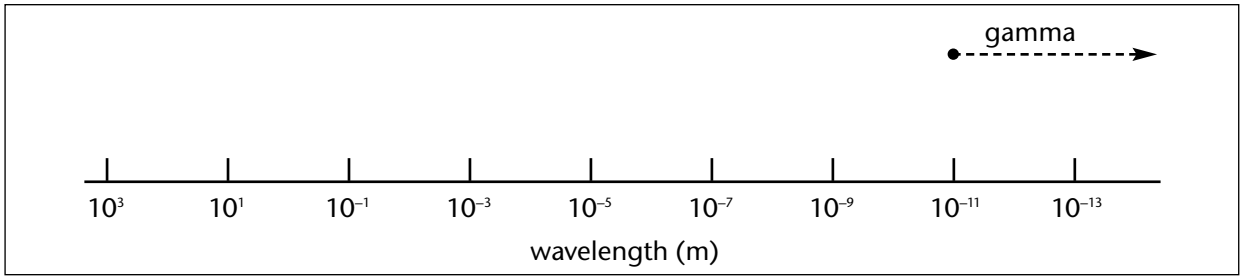
Gamma rays, infra-red waves, light rays, microwaves, radio waves, ultra-violet rays and X-rays are similar. They all travel at the speed of light (300 000 000 m/s or 3×10^8 ms^{-1}) and they all have an electric and a magnetic wave component. However, they are different because they vary in wavelength, frequency and energy. They are also produced by different processes and have different effects on us and other things.

1 Here is the list again. This time the radiation list shows the wavelength of each radiation type: gamma rays ($< 1 \times 10^{-11}$ m), infra-red waves ($7 \times 10^{-7} - 1 \times 10^{-3}$ m), light ($4 \times 10^{-7} - 7 \times 10^{-7}$ m), microwaves ($1 \times 10^{-3} - 1 \times 10^{-1}$ m), radio waves ($> 1 \times 10^{-1}$ m), ultra-violet rays ($1 \times 10^{-8} - 4 \times 10^{-7}$ m) and X-rays ($1 \times 10^{-11} - 1 \times 10^{-8}$ m). Rewrite the list. Begin with the longest wavelength.

Longest _____

_____ shortest

2 The diagram below shows a wavelength scale. Draw short, labelled lines like that shown for the gamma ray to indicate the wavelength range of the other electromagnetic (EM) waves.



3 Wave frequency is measured in hertz. A frequency of one hertz (1 Hz) means one cycle of the wave in one second. The list of electromagnetic waves is given again. This time their frequency ranges are shown: gamma rays ($> 3 \times 10^{19}$ Hz), infra-red waves ($3 \times 10^{11} - 4 \times 10^{14}$ Hz), light ($4 \times 10^{14} - 7 \times 10^{14}$ Hz), microwaves ($3 \times 10^{9} - 3 \times 10^{11}$ Hz), radio waves ($< 3 \times 10^{9}$ Hz), ultra-violet rays ($7 \times 10^{14} - 3 \times 10^{16}$ Hz) and X-rays ($3 \times 10^{16} - 3 \times 10^{19}$ Hz). Rewrite the list beginning with the smallest frequency.

Smallest _____

_____ highest

4 Compare the two lists. What do you notice?

_____

5 One joule (1 J) is a unit of energy equal to the work done when a force of one newton (1 N) acts through a distance of one metre (1 m). It's the work you do when you lift a 100 g bar of chocolate from your trouser pocket to your mouth. The energy ranges of the different EM radiations are listed after each radiation type: gamma rays ($> 2 \times 10^{-14}$ J), infra-red waves ($2 \times 10^{-22} - 3 \times 10^{-19}$ J), light ($3 \times 10^{-19} - 5 \times 10^{-19}$ J), microwaves ($2 \times 10^{-24} - 2 \times 10^{-22}$ J), radio waves ($< 2 \times 10^{-24}$ J), ultra-violet rays ($5 \times 10^{-19} - 2 \times 10^{-17}$ J) and X-rays ($2 \times 10^{-17} - 2 \times 10^{-14}$ J). Rewrite the list, ordering it by energy level.

Smallest _____

_____ highest

6 How does this list compare with your other lists?

_____

7 Consider this object list: atom, bee, house, molecule, nucleus (of an atom), pinhead, plant cell. Match this list with the ordered wavelength list of EM radiation. (Question 1.)

Radiation list: _____

Object list: _____

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The electromagnetic spectrum

3b

8 The following list states sources of the various types of EM radiation: electric radiator, light bulb, microwave oven, nuclear reactor, quartz lamp, radio transmitter, X-ray machine. Match this list with the ordered wavelength list of EM radiation.

Radiation list: _____

Source list: _____

9 The following list indicates possible outcomes or uses of the various types of EM radiation: communication, cooking, death, deformities, heat, skin cancer, sight. Match this list with the ordered wavelength list of EM radiation.

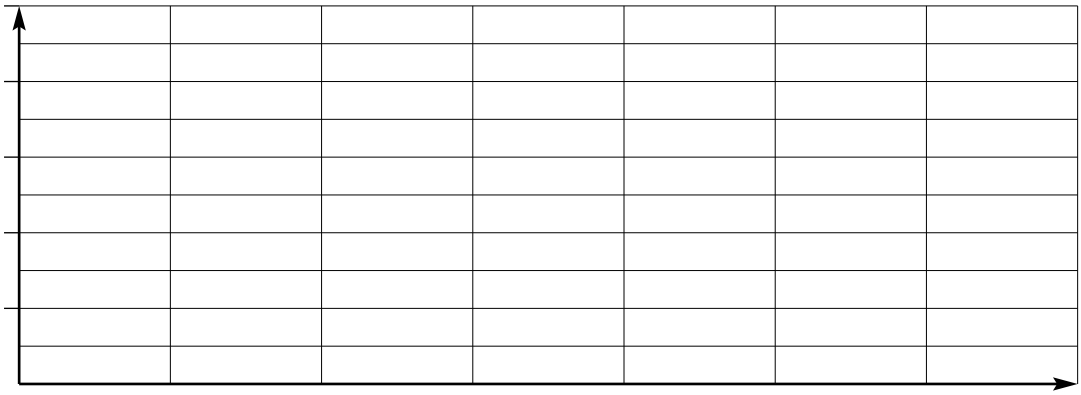
Radiation list: _____

Outcome list: _____

10 Much solar radiation is trapped by the earth's atmosphere. The table records the average atmospheric heights at which various types of electromagnetic radiation are mostly absorbed.

Radiation type	**Gamma**	**Infra-red**	**Light**	**Microwave**	**Radio** waves	X-rays	Ultra-violet
Absorption altitude (km)	25	12	0	25	0	100	30

Show this information in a column graph. List the radiation types in order of wavelength on the x-axis.



Note: Some ultra-violet rays as well as some microwaves do reach the surface of the earth.

11 Cross out the incorrect term from each set of brackets.

The ozone layer in the (atmosfere / atmosphere) is a thin, low altitude (laier / layer) of ozone gas, O_3, a form of oxygen. It is able to (absorb / absorp) ultra-violet rays. However, the (ozone / ozoone) layer has thinned out (during / while) the last few (decades / decads). As a result during spring a large ozone hole (developes / develops) above Antarctica and a smaller (one / own) above the Arctic. These holes (rebuild / rebuilt) again over summer.

(Unfortunately / Unfortunatly) the ozone layer is also somewhat (fin / thin) above southern parts of (Australia / Austria), especially during spring and (earlie / early) summer. (Sunscrean / Sunscreen), hats, sunglasses and long-sleeved (clotheing / clothing) offer much (protection / protektion) against these (dangerous / harmless) ultra-violet rays. Chlorofluorocarbons and halons are (beleaved / believed) to have caused the (los / loss) of ozone. Cold temperatures (increase / increese) the speed of ozone (depleetion / depletion). If the (hole / whole) above the Arctic were to (increase / increese), large (people / populations) in northern Europe, northern (asia / Asia) and northern North America (could / would) be threatened (by / with) melanomas caused by (ultra-violet / ultrared) radiation.

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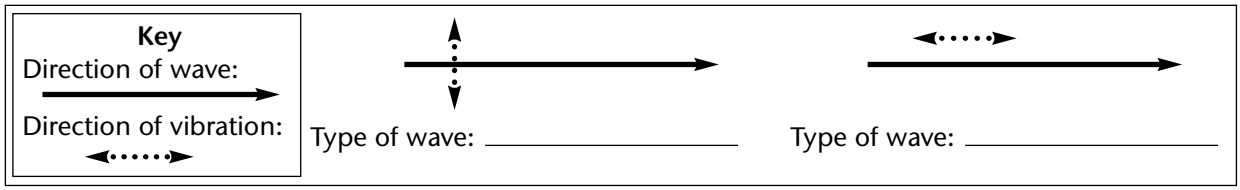
Types of waves

4a

Waves involve vibrations. They carry energy from one place to another. There are two types of waves: longitudinal (lengthwise) waves in which the medium carrying the wave vibrates along the direction in which the wave is travelling, and transverse (across) waves in which the vibrations are perpendicular to the direction of wave movement. Longitudinal waves are sometimes called compression waves.

1 Use the spaces provided to label each wave diagram.



2 One type of wave (sound waves are of this type) always consists of vibrating particles that pass energy on to other particles, which then start to vibrate themselves. State which of the above wave types could be a sound wave. Justify your answer.

_____

3 If the energy of vibrating particles is passed on from particle to particle, in which medium would sound travel faster: air, water or iron? Choose a medium and justify your answer.

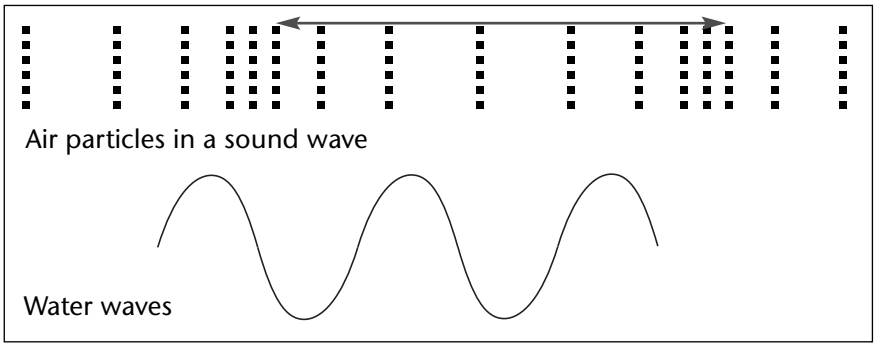
_____

4 Suggest what happens to the speed of sound at higher levels in the earth's atmosphere. Justify your answer.

_____

5 Explain why sound does not travel in space. _____

6 In addition to vibration patterns, waves can differ in wavelength and frequency. Generated by vibrating sources, waves will have the same frequency as their source and the pattern of disturbance will repeat itself regularly. A wavelength is the distance between any two closest points along the wave for which the disturbances are in step. Indicate three wavelengths on each diagram at right. One wavelength has been drawn as an example.



7 **a** Amplitude refers to the range of movement of the vibration in a wave. It is about loudness or intensity. In longitudinal waves the amplitude is measured from the centre of a compression to the centre of the closest rarefaction. Circle and label the middle of a compression and the middle of the rarefaction of the sound wave above.

b Measure the amplitude of that sound wave in mm. _____

8 **a** In transverse waves the amplitude goes from the zero line (i.e. the line around which the disturbance vibrates) to the top of a crest (or bottom of a trough). Circle and label a crest and a trough in the transverse wave above. Then draw the zero line.

b Measure the amplitude of the water wave in mm. _____

The frequency of a wave measures the number of cycles of the disturbance or the number of vibrations that take place in one second. For a travelling wave, the number of wavelengths that pass a point in one second is also a measure of the frequency. The unit of frequency is the hertz; e.g. 2 Hz (i.e. two hertz) is two cycles per second. Frequency is calculated by dividing the number of cycles or waves by the time they took to travel past one point (i.e. 2 waves/1 s = 2 Hz). (In general, if the wavelength is big, then the

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Types of waves

4b

frequency is _____.)

9 Calculate the frequencies of the musical notes listed in the table below. (When talking about music and the human ear the word pitch is often used rather than frequency.) Note A above the middle C is known as concert pitch.

Note	Number of cycles	Time (s)	Calculation	Frequency (Hz)
Middle C	256	1		
A	880	2		
D	864	3		
High C	2048	4		
E	1400	5		
F	341	1		
G	678	2		
A	660	3		
B	960	4		
A	550	5		
B	960	2		

10 List the notes in order of frequency (low to high) and select eight consecutive notes to form an octave. Start with A (220 Hz) and list the immediately following notes in order.

_____

11 List the frequencies of all A notes.

_____

12 Write a generalisation about the relationship between any two adjoining A notes.

_____

13 The speed of a wave depends on the medium it is travelling through. Light travels through empty space (i.e. a vacuum) at 300 000 000 m/s (more accurately 299 792 458 m/s), through glass at 200 000 000 m/s, through diamond at 125 000 000 m/s and through water at 225 000 000 m/s. Light speed through air is practically the same as light speed in a vacuum. Record these speeds in the table (highest to lowest).

Medium				
Speed of light				

14 Sound travels through air at 20°C at a speed of 342 m/s. This speed is about one million times slower than the speed of light. When air temperature drops to 0°C, sound speed drops to 331 m/s. Water carries sound at a speed of 1 500 m/s, brick at 3 600 m/s, wood at 4 000 m/s and iron at 6 000 m/s. Suggest a reason why sound speed drops, when the air temperature drops. (Think of the kinetic particle theory.)

_____

15 Make a generalisation about how the speed of sound changes, as the density of the transmitting medium changes.

_____

16 Relate the speed of light to the density of the transmitting medium.

_____

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Speed of light and sound

5a

The speed of waves varies depending on the medium (i.e. substrate, substance) the wave is travelling through. For light waves, denser mediums slow down the waves. For sound waves, denser mediums increase the speed. Sound waves must travel through a medium. A vibrating particle in this medium bumps into another particle, which then also starts to vibrate and so bumps into yet another particle. The kinetic particle theory can be used to explain the speed of sound in different mediums.

1 Explain how an iron bar conducts sound faster than air does.
(Hint: Think of the proximity of particles.)

2 Predict and explain whether cold air particles (molecules) will transmit sound at a higher speed than hot air particles. (Hint: Think of the speed of particles.)

Light waves are not dependent on the presence of a medium. Their electromagnetic vibrations can travel through a vacuum as well as a medium. However, when light waves spread through a medium, their energy is absorbed by the atoms of the medium and then given out again.

3 Explain why light travels more slowly in glass than it does in air.

4 Explain why air allows light to travel through it at nearly the same speed as in a vacuum?

5 Light travels at a speed of 300 000 000 m/s and sound at 333 m/s.

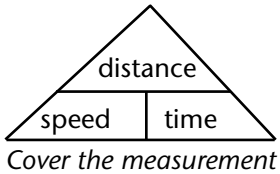
a Convert these speeds into km/s. (Hint: 1 m equals 1/1000 km.)

b Now convert these speeds into km/h. (Hint: 1 s equals 1/3600 h.)

6 Timekeepers at the finish line of a race are told to watch the smoke of the starting gun rather than listen for the sound. Explain.

7 A plane in the air looks further forward than it sounds. Explain.

8 Six km away fireworks are shot into the air. Calculate how soon you are able to see and hear them. (Hint: Recall the formula for speed)



Cover the measurement you want with your finger

9 During a thunderstorm you see a lightning flash and three seconds later you hear the thunder. How far away is the thunderstorm?

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Speed of light and sound

5b

10 While engaged in intergalactic space travel, a starship which is about 3 000 000 km away from you explodes. Calculate how quickly you would see and hear that explosion.

_____

_____

11 The speed of a wave is calculated by multiplying its wavelength by its frequency. State units for these three quantities:

speed: _____ wavelength: _____ frequency: _____

12 **a** Write the formula for the speed calculation:

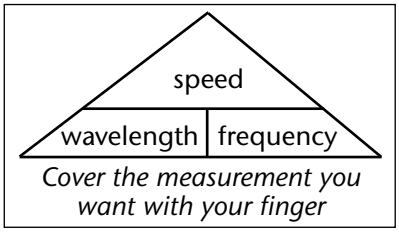
_____

b Adjust the formula: make wavelength the subject.

_____

c Adjust again: make frequency the subject.

_____



13 The following table shows information about musical notes. Complete the table using the speed formula or any of its adjustments. Show your working in the Calculation column.

	Frequency (Hz)	Speed at 20°C (m/s)	Calculation	Wavelength (m)
Middle C	256	342		
High C	512	342		
Low C	128	342		
Concert pitch (middle A)	440	342		
Lowest heard sound	20	342		
Highest heard sound	20 000	342		

14 The speed of light has been given the symbol c; it is an important number. c is also the speed of all electromagnetic radiation waves. State the speed of light in m/s and in km/s.

_____

15 The following table shows the frequencies and wavelengths of several electromagnet waves. Complete it by referring to the formulas in question 12.
(Hint: The speed of light is 300 000 000 m/s or 3×10^8 m/s.)

Wave type	Frequency (Hz)	Wavelength (m)	Calculation
Gamma rays	$3 . 10^{19}$		
Red light		$750 . 10^{-9}$	
Blue light	$6.7 . 10^{14}$		
Radio waves		10^2	

Note: To multiply two exponential numbers, add their exponents: e.g. $10^3 . 10^{-2} = 10^{3+(-2)} = 10^{3-2} = 10^1$ (i.e. 1000 . 0.01 = 10). To divide two exponential numbers, subtract the exponents: e.g. $10^5 / 10^3 = 10^{5-3} = 10^2$ (i.e. 100 000 / 1 000 = 100); $10^2 / 10^{-3} = 10^{2-(-3)} = 10^{2+3} = 10^5$. Numbers without exponents are treated in the usual way: e.g. $6 . 10^6 / 3 . 10^3 = 6 / 3 . 10^{6-3} = 2 . 10^3$.

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A brief history: communication events

6a

Some events in the history of communication are recorded in the following boxes:

1956: Channel 9 started broadcasting, Australia ◯	1923: Wireless radio started broadcasting, Australia ◯	1838: Samuel Morse invented telegraph, USA ◯
1880: Telephone exchange opened in Melbourne, Australia ◯	1968: Australia linked to global satellite system ◯	1986: Optical fibre cable layed between Sydney and Melbourne ◯
1990: Phonecards introduced by Telstra, Australia ◯	1993: Digital mobile phones introduced, Australia ◯	1920: Radio broadcasts of voices began ◯
620: Chinese court circular called a pao appeared, China ◯	1876: Alexander Bell invented short distance telephone, USA ◯	1892: Automatic telephone exchange operated, USA ◯
1890: Heinrich Hertz built transmitter and receiver, Germany ◯	1998: Guglielmo Marconi sent radio morse code message, Italy ◯	60 BC: Roman gazette Acte Diurna displayed in public places, Italy ◯
1926: John Baird showed black and white television, Scotland ◯	1456: Johann Gutenberg printed books with movable type, Germany ◯	1795: Semaphore system (= flags on posts) took 2 min for 240 km, France ◯
1948: Vinyl record used to record music ◯	1615: Newspaper 'Frankfurter Zeitung' published, Germany ◯	750: Books were printed, China ◯
5000 BC: Message sticks carried simple information, Australia ◯	7000 BC: Drum beats conveyed short messages across jungle, Africa ◯	105: Tshai Lun write on paper made from bark and bamboo, China ◯
1084 BC: Fall of Troy news relayed by chain of fires to Greece ◯	2000 BC: Aztecs, Babylonians, Chinese, Egyptians wrote in pictures ◯	1889: Almon Strowger patented dial telephone system, USA ◯
1464: State post delivered mainly official mail, France ◯	1939: John Baird showed colour television, Scotland ◯	1935: Tape recorder shown in Berlin, Germany ◯

1 Order the events mentioned above. Write '1' in the circle of the earliest, then '2' in the circle of the next event and so on.

2 On the back of this sheet or on a separate sheet of paper, draw a timeline showing the events mentioned above. Add any other events or devices you can think of.

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A brief history: communication events

6b

3 Suggest two differences between early and more recent communication devices.

4 Suggest another difference if you can.

5 State one early and one late exception to your generalised differences.

6 For each of the properties listed in the table below, list three communication devices that have that property.

Reaches many people	Reaches only one person	Travels very fast	Can travel a great distance	Can carry much information

7 Choose one of the devices mentioned in the brief history of communication events. Imagine there is a person who has never heard of this device. Write a procedure telling that person how to use it.

8 Select a long distance communication device and explain how it might send your word to a receiver.

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Speaking and listening with a telephone

7a

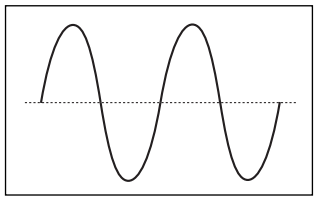
Early telephones contained a carbon microphone in a handheld mouthpiece along with a separate earpiece, dialing mechanism and other parts. The microphone changed sound waves into varying electric currents. The earpiece converted varying electric currents from another telephone back into sound waves.

High pitch sounds have high frequencies and result in electric currents that change direction (alternate) quickly (with high frequency). Deep or low pitch sounds with their lower frequencies end up as electric currents with a lower frequency. Soft sounds have vibrations with small amplitudes, while loud sounds have vibrations with large amplitudes.

A carbon microphone is a small box with two metal plates – one fixed and the other free to vibrate, each connected by wire with other parts of an electrical circuit. Carbon powder fills the space between the two plates. When the carbon powder is loosely packed, it conducts electricity poorly. When it is compressed, it is a better conductor of electricity.

A soft, deep sound hits the free-moving plate (diaphragm) of a carbon microphone resulting in weak and low frequency vibrations of the plate. These plate vibrations cause weak and low frequency electricity pulses in the wires leading away from the carbon microphone.

The diagram at right represents a medium pitched normal sound. The sound wave has been represented like a transverse wave in the diagram, although it is really a longitudinal wave. Transverse waves are easier to draw and differences can be shown clearly, so particular sound waves are often sketched like transverse waves.



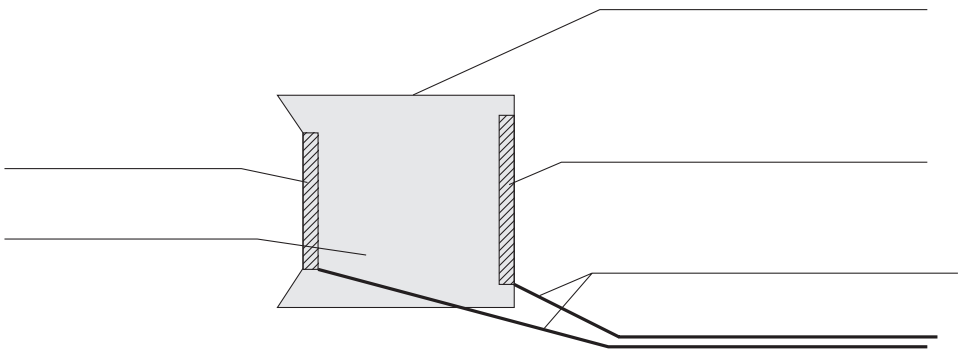
1 Draw wave diagrams in the cells of the following table. Use the diagram above as a referent for amplitude and frequency.

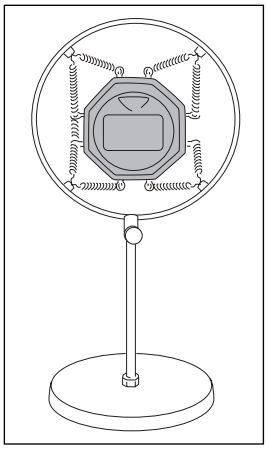
Higher pitch	Lower pitch	Quieter sound
Louder sound	Higher, louder	Deeper, quieter

2 Complete the table by describing the characteristics of particular sounds and their resulting electric current. Choose from higher or lower, higher or lower, stronger or weaker, and higher or lower.

Sound	Sound wave frequency	Sound wave amplitude	Electric pulse strength	Electric pulse frequency
Soft and deep				
Soft and high				
Loud and high				
Loud and deep				

Carbon microphone





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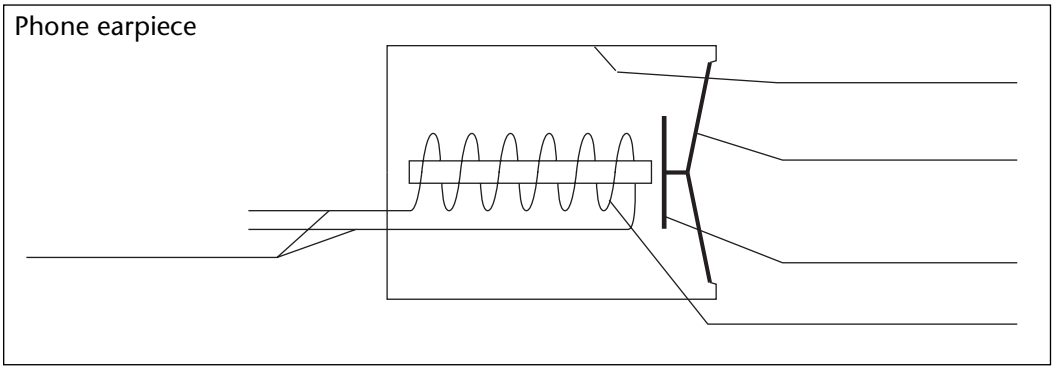
Speaking and listening with a telephone

7b

3 Label the diagram of the carbon microphone on the previous page.



Fact 1: A strong alternating current in the wire around the iron core causes a strong varying magnetic field, particularly around the end of the iron core.

Fact 2: A strong varying electromagnetic field attracts and repels the iron plate strongly resulting in large amplitude vibrations of the plate.

Fact 3: A large amplitude movement of the diaphragm attached to the plate (usually a piece of cardboard) produces a loud sound.

Fact 4: A high frequency electric pulse results in a high frequency movement of the diaphragm and this causes a high pitch sound.

4 Label the diagram of the earpiece shown above.

5 Write a fact list for a weak, low frequency alternating current entering the earpiece.

6 Imagine you are building a telephone system made up of two phones – so there are two earpieces, two microphones, two batteries and several wires. Sketch how you would arrange these pieces. (Hint: There are actually two circuits.) Note that in mobile phones, the connecting wires have been replaced by radiowaves.

7 A high pitch sound goes from your microphone to the earpiece of a friend's phone. On a separate sheet of paper, write a paragraph or draw a flowchart explaining how this sound is transferred.

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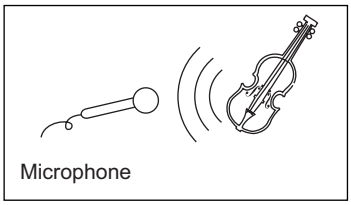
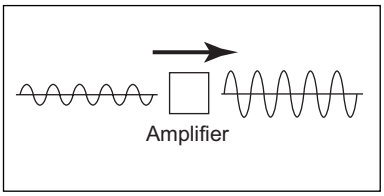
Sending and receiving music by radio

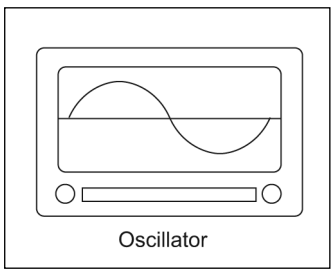
8a

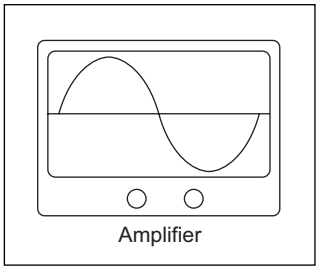
Studio band music is collected by microphones, manipulated and eventually the changed signal is transmitted from the radio station's antenna. Far away (or near by), a radio receiver picks up those radio waves and uses them to reproduce the studio music. Some details of the steps in the process are described below.

1 Match the writing about a radio transmitter to the ordered diagrams by drawing lines to connect them.









A
An oscillator produces high, constant frequency electric vibrations called a carrier wave. For example, ABC Sydney on the AM band has a carrier wave frequency of 702 kHz.

B
Sound waves are changed into varying low frequency electric impulses by a microphone. The frequency and strength of the impulses depend on the pitch (i.e. frequency) and loudness (amplitude) of the sounds.

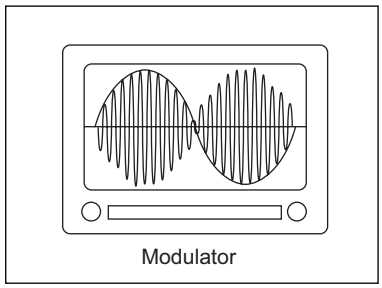
C
The audio wave and carrier wave are merged into a high voltage modulated wave by a modulator.

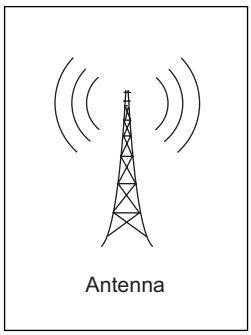
D
The carrier wave is amplified by an intermediate amplifier.

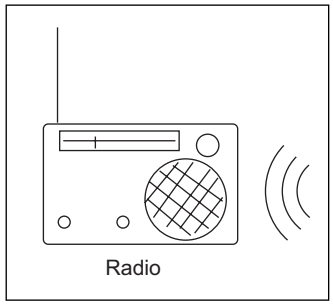
E
The electric impulses, forming the audio wave, are strengthened by an amplifier.

F
The high voltage, high frequency modulated waves produce radio waves through an antenna. The waves travel at 300 000 km/s away from the antenna.

G
The radio station JJJ on the FM band has a carrier wave frequency of 105.7 MHz in Sydney. It is these waves that are collected by a radio and changed into sound.







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Sending and receiving music by radio

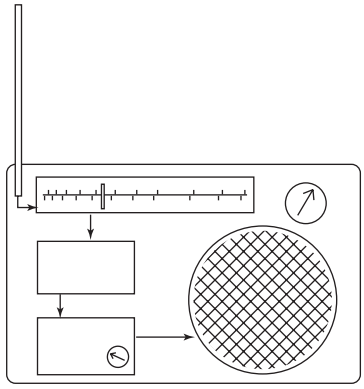
8b

2 Refer to the previous page. Write an explanation for how a radio station changes music into radio waves. Do not forget to write introductory and concluding paragraphs. Using time and cause conjunctions such as when, after, then, because, since and therefore will help the clarity of your explanation.

3 The sentences below describe the operation of a radio receiver. Use the sentence letters to label the diagram showing the part that performs the operation described in the sentence.

Letters	Sentences	Order
A	An alternating current of a particular frequency is selected by the **tuner** and passed on.	
B	In the **demodulator** the carrier wave is separated from the signal it was carrying. Only the audio wave is passed on.	
C	Radio waves are picked up by the **antenna**, which generates weak alternating currents.	
D	The amplified audio wave is converted into sound and music by the **speaker**.	
E	The audio wave is amplified to a desired loudness by the **amplifier**.	

Radio receiver



4 Order the sentences (starting with the antenna) to show the change from radio wave to sound wave by writing 1, 2, etc. in the Order column of the table above.

5 Label the diagram of the radio above using the bold terms in the sentences.

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Sending and receiving music by radio

8c

6 The sentences in question 3 begin with 'waves' or 'current' because the focus is on waves and currents. The sentences are all in passive voice. Change them to active voice by beginning the sentences with the bold terms. Write the sentences in your numbered order.

7 Use information from the drawing and the sentences to write an explanation of how a radio produces music from radio waves. Draw and label a diagram in the space below to illustrate your explanation.

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Reflection and refraction – A contrast

9a

To contrast means to describe the differences between things. When we contrast information, it is generally helpful to use contrasting conjunctions such as *more than, less than, different from, in contrast to* and *rather.*

1 Can you think of other contrasting words? Write them on the line below.

Using these words or phrases in a sentence highlights differences. Differences between a concave and a convex mirror are indicated in each of the following sentences:

- A concave mirror curves inwards, while a convex mirror curves outwards.
- In contrast to concave mirrors, which curve inwards, convex mirrors curve outwards.

2 Write two contrasting sentences about mirrors using the information given in this table:

Feature	Concave mirror	Convex mirror
Mirror sketches		
Focus position	In front of mirror	Behind mirror
Image of close object	Upright, magnified	Upright, diminished
Image of distant object	Upside-down, diminished	Upright, diminished

3 Write two contrasting sentences about lenses using the information given in this table:

Feature	Concave lens	Convex lens
Sketch		
Image	Diminished	Magnified

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Reflection and refraction – A contrast

9b

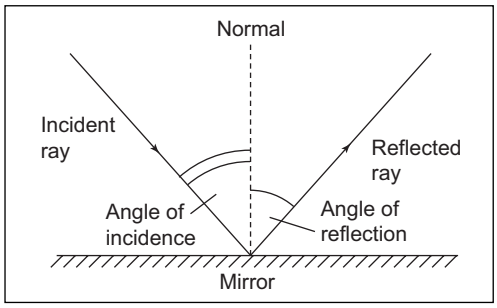
Reflection

Definition: Light that bounces off surfaces is reflected light.

Observation: Mirrors reflect light orderly and images are formed. Surfaces that are rough reflect light in all directions to giving overlapping images.

Law: The angle of reflection is equal to the angle of incidence.

Sketch:



Refraction

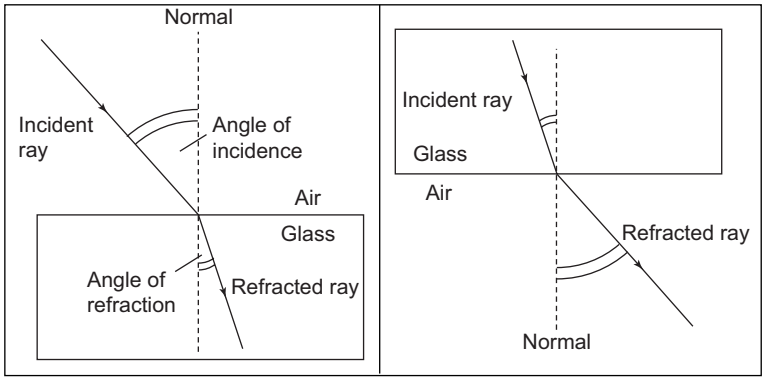
Definition: Light that bends as it enters another medium is refracted light

Observation: Smooth lenses make images look bigger or smaller (magnified with convex lenses, diminished with concave lenses).

Law:

- Light is bent towards the normal as it enters a denser medium, e.g. travels from air to glass.
- Light is bent away from the normal as it enters a less dense medium, e.g. as it passes from glass to air.

Sketches:



4 Look at the diagrams above and then define the following:

a Incident ray: _____

b Reflected ray: _____

c Refracted ray: _____

d Normal: _____

e Angle of incidence: _____

f Angle of reflection: _____

g Angle of refraction: _____

5 Look at the table about reflection and refraction, and then write five contrasting sentences.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

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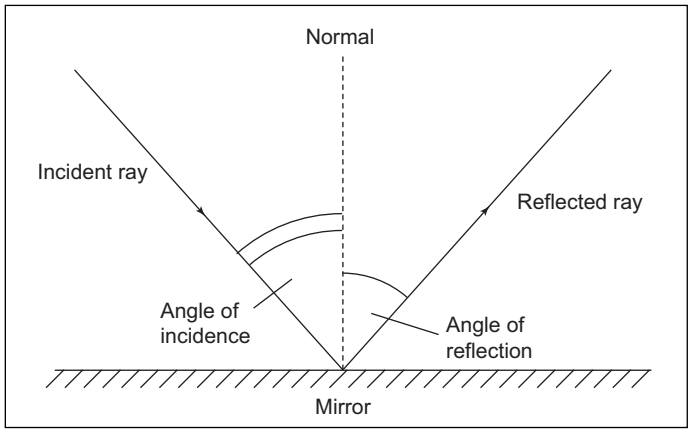
Reflection, refraction, diffraction, scattering 10a

Sound and light carry messages from one place to another. Light waves are transverse waves that do not require a medium for transmission. Sound waves are longitudinal waves that do require a medium for transmission. Sound and light waves seem to travel in straight lines ... or do they?

Reflection

Like a bouncing ball, light can bounce off a surface. The reflected light changes direction when it bounces off a surface. Measurements show that the angle of incidence equals the angle of reflection (the law of reflection).



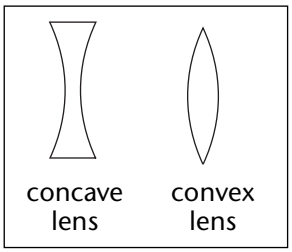
Smooth surfaces act as mirrors – light rays are reflected uniformly and form images. Each bit of a rough surface also obeys the law of reflection, but the overall effect of the uneven surface is to scatter the light everywhere. Clear images are not formed.

Optical fibres are tubes with an extremely thin glass core. They are produced so that light will pass along the fibres bouncing off the walls (internal reflection). As (ideally) no light leaves through the sides, a light ray is able to travel large distances in the fibre.

Sound is also reflected. We detect echoes when a reflected sound wave reaches our ears some time after the original sound wave was heard. Reverberations are sounds that bounce off surfaces in all directions and cause annoying background noise.

Refraction

During refraction, light and sound waves change direction as if travelling around a corner. When waves pass from one medium into another, the path of the waves bends. There is also a corresponding change in the speed and the wavelength of the waves.

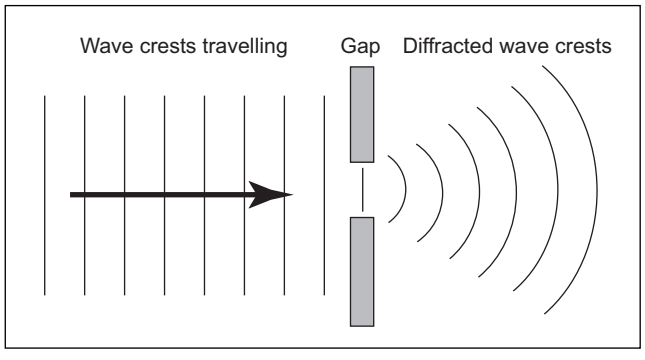


When light enters glass from the air, it bends one way; when it travels from glass to air, light bends the other way. Objects appear to be bigger when seen through a convex lens and smaller when seen through a concave lens.

Hot air and cool air have different densities and so form different mediums optically. As a result, light rays curve when they travel from hot to cool air.

Diffraction

Diffraction also involves a change in direction, and occurs when waves pass through small openings or around a barrier in the wave path. During diffraction through a slit the waves spread out as if the slit or gap acts as a new point source for the waves.



Diffraction

Scattering

During scattering light is deflected from its path in all directions. Atmospheric scattering is caused by deflection from particles in the atmosphere. As a result of scattering distant objects may appear red or blue, or other colours.

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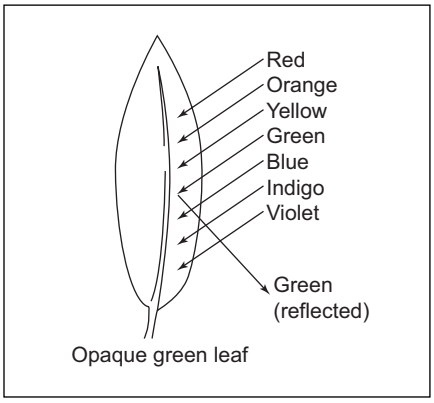
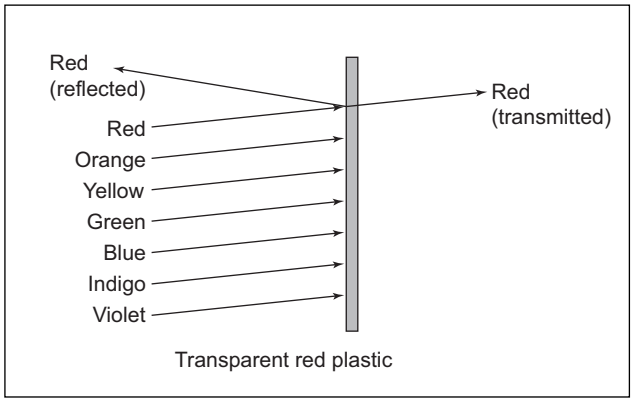
Reflection, refraction, diffraction, scattering 10b

Reflection, absorption and transmission

White light is a mixture of the seven rainbow colours red, orange, yellow, green, blue, indigo and violet (ROY G BIV). Opaque materials reflect all or some colours, while the other colours are absorbed. The reflected colours are responsible for the material's colour. An opaque object lets no light travel through it. However, a transparent material transmits and reflects some colours, while the other colours are absorbed. Again the reflected and transmitted colours produce the colour of the material.





Below each of the following statements and/or sketches write the appropriate term (diffraction, reflection, refraction, scattering, transmission) and the applicable term (light, sound, water).

1 **a** sun ⟶ earth

_____ _____

b person ⟶ person

_____ _____

2
- The sun looks red in the evening.
- The sun looks white during the day.
- The sun looks orange in the morning.

_____ _____

3 A clear cloudless daytime sky is blue, because molecules in the air scatter blue light from the sun more than they scatter red light.

_____ _____

4 A leaf looks green, because green light is reflected while the other rainbow colours are absorbed by the leaf.

_____ _____

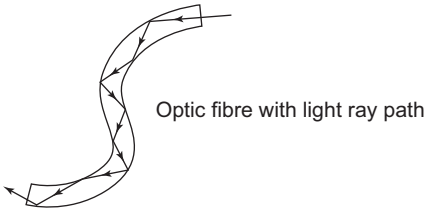
5 A piece of cardboard looks blue because all other rainbow colours are absorbed.

_____ _____

6 A shout against a distant mountain cliff returns to you a little later as an echo.

_____ _____

7 Light pulses bounce around inside an optic fibre without leaving it and so are able to travel large distances.



_____ _____

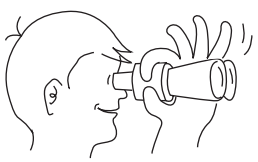
8 An image of a tree can be seen on a polished car door.

_____ _____

9 Atmospheric mirages are caused by light passing through and being refracted by layers of air that have different temperatures and therefore different densities.

_____ _____

10 Binoculars, microscopes and reading glasses make objects appear bigger.



_____

_____

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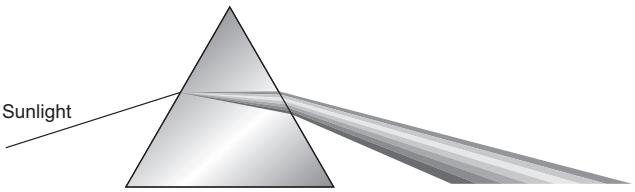
Reflection, refraction, diffraction, scattering 10c

11 Distant mountains appear blue.

12 During a sunset or sunrise the blue colours are scattered out of the sunlight leaving only reddish colours.

13 Newton's famous experiment with sunlight and a triangular prism showed that white light is made up of the beautiful rainbow colours (ROY G BIV).

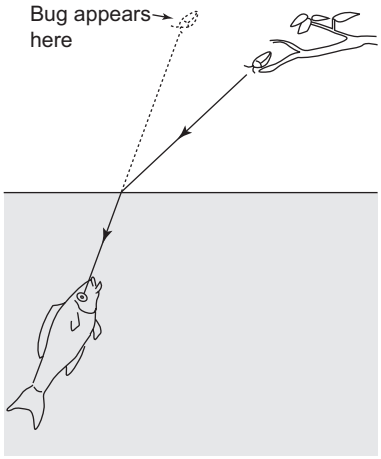


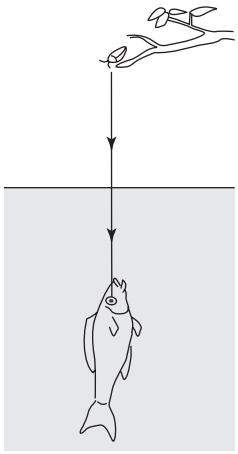
14 Shouting inside a concrete building makes it difficult to hear anything.

15 Sunlight on a piece of paper makes the paper appear white without producing an image.

16 The air directly above water is cooler than the air far above water. Since sound waves travel more slowly in cooler air than they do in warmer air, the portion of the wavefront directly above the lake is slowed down, while the portion of the wavefronts far above the water speeds ahead. Subsequently, the direction of the surface wave changes, refracting downwards towards the water and sound seems to follow the curvature of the sea.

17 The archer fish knows to be directly under a bug if it wants to catch it. When looking at an angle, the fish can not be sure where the insect is.





18 We hear people speaking from adjacent rooms when the door has been left ajar.

19 When water waves enter through the narrow harbour opening, they spread out across the whole harbour and make all boats rock.

20 Your shoes appear black because most rainbow colours are absorbed; hardly any light is reflected.

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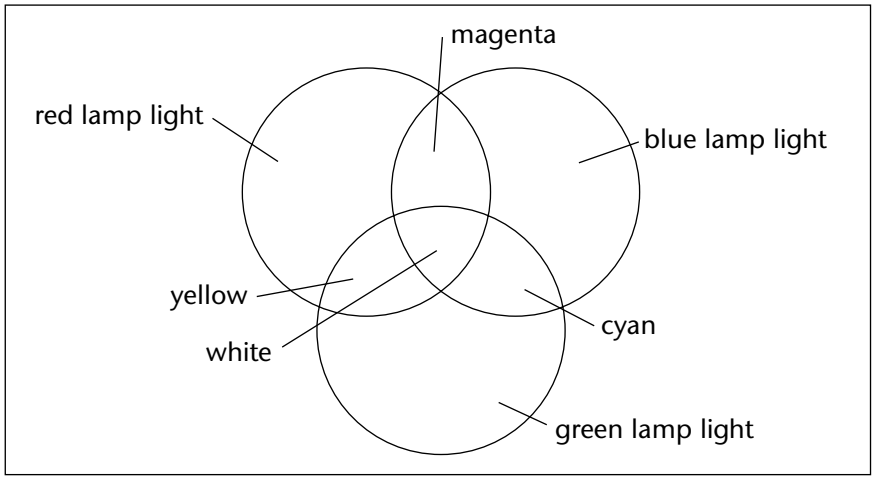
Whose colour?

11a

1 Cross out the incorrect term from each set of brackets.

Let the stage light operator speak:

Lights of various (colours / colors) are shone onto the stage to (creat / create) certain (affects / effects). Often coloured (lights / lites) are mixed by (shining / shineing) them on the same (spot / spots). When red and (blu / blue) lights are mixed, a magenta colour, which is dark red, is (produced / producted); green and red make (yello / yellow); and blue and green give cyan (light blue); and all (free / three) colours (to gether / together) produce white light.

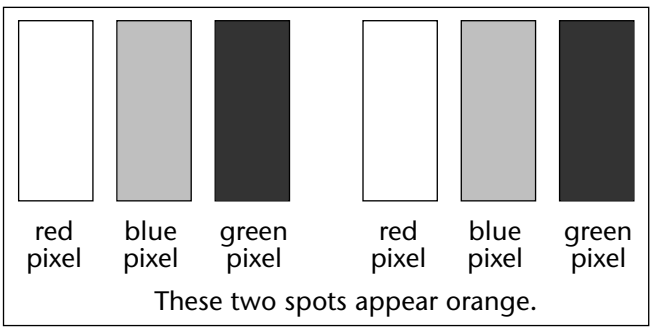


2 Cross out the incorrect tense from each set of brackets.

Let the television engineer speak:

Television screens and other colour monitors (are made up / were made up of) of thousands or even millions of pixels, tiny coloured dots or stripes. These pixels (are stimulated / will be stimulated) and then (glow / glowed) for a very short time. Three primary colours (are used / were used): red, green and blue. From these all other colours (can be produced / have been produced)). Three separate electron guns in a TV picture tube (are stimulating /stimulate) each colour pixel. Flat screens (had /have) bubbles which (are stimulated / were stimulated). Because the coloured dots (are / were) too small to be seen individually, they (merge / merged) to produce one mixed colour spot. If a group of red and green dots (are lit up / will be lit up), yellowish spots (result / results). But if the green dots (are / will be) not so bright, the coloured spots (were appeared / would appear) orange.

You (have been able to see / might be able to see) the colour pixels, if you (look / will look) at a TV screen from very close. A computer monitor (had / has) much smaller pixels (having resulted / resulting) in a sharper picture.



3 Cross out the incorrect preposition from each set of brackets.

Let the painter speak:

A painter doesn't use light (from / of) dots or lamps. A painter mixes coloured pastes and relies (by / on) the colours which are reflected (for / from) those pastes. Magenta paint mixed (on / with) yellow paint gives a red colour; and cyan and magenta produce a blue colour. All colours mixed together end up black. The three paints cyan, magenta and yellow produce the greatest range of colours when mixed.

See diagram, next page.

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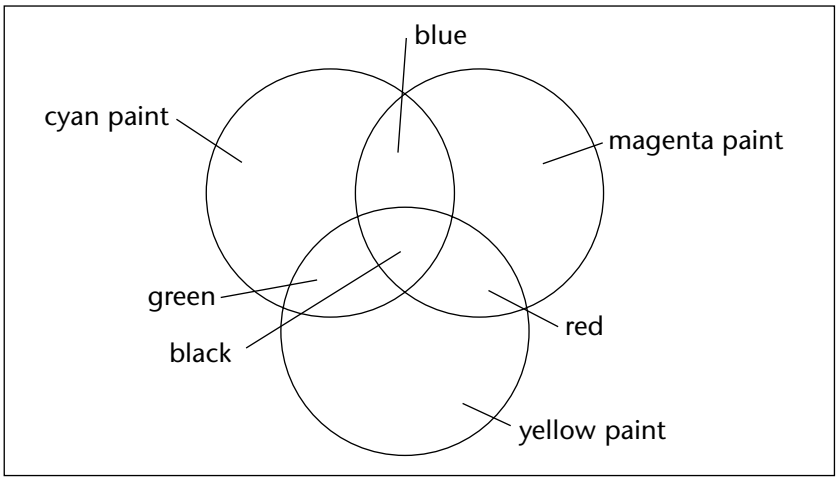
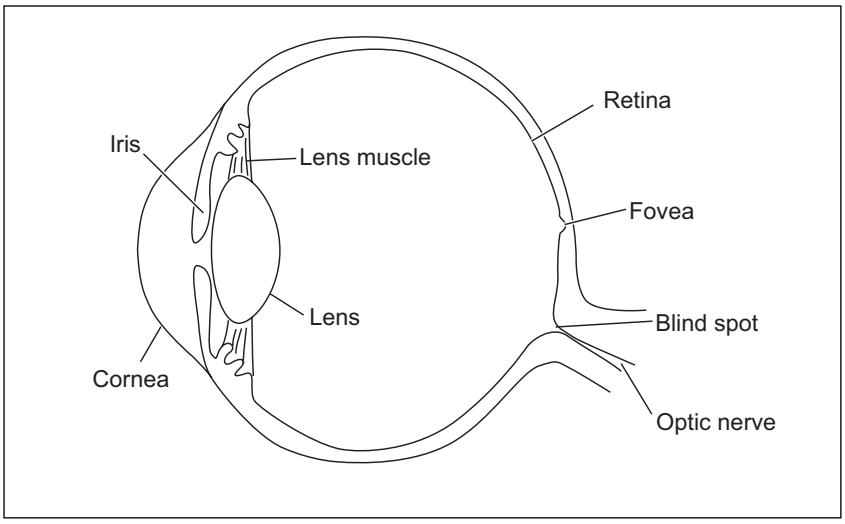
Whose colour?



A painter uses colour to create mood, shape, perspective, depth, contrast and harmony. Very subtle changes (in / on) hue and colour result (in / on) interesting effects.

4 Order the sentences in each of the three paragraphs by writing numbers in front of the sentences.

Let the doctor speak:



Paragraph 1:

☐ **a** Light stimulates nerves in the eye that send a message to the brain.

☐ **b** Then the brain produces a picture.

Paragraph 2:

☐ a On the retina light sensitive nerve endings are stimulated: When it is dark, rods are stimulated to produce a black and white picture; when it is bright, three types of cones (i.e. red, blue and green cones) are stimulated to varying degrees and produce a colour picture in our brain.

☐ **b** The brain interprets them and builds up a picture.

☐ **c** The stimulated nerve endings, i.e. the rods and cones, send electric impulses to the brain.

☐ **d** When light enters the eye, the cornea and lens focus the light rays onto the retina (usually near the spot called the fovea).

(continued next page)

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Whose colour?

Paragraph 3:

- ☐ **a** Because the brain and not the eye develops the picture, it is possible to see something which is not even there.
- ☐ **b** However, despite the existence of your blind spot, when you normally look around, there is no hole or gap in the pictures that you see.
- ☐ **c** It is the brain that compensates for the hole due to the blind spot and so completes the image in the brain.
- ☐ **d** Right then the image of the 'X' falls on the blind spot in your eye (there are no rods and cones where the optic nerve leaves the eye ball) and so you cannot see the 'X'.
- ☐ **e** When you look at the dot below with only your left eye and move the paper slowly backwards and forwards, at some point (when the page is about 15 cm from your eye) you will no longer be able to see the 'X'.
- ☐ **f** X ●

5 Summarise the information by completing the table as much as possible.

Let the biologist speak:

It is difficult to know exactly what other animals see. Many animals do not see like humans, who have three types of cones in their retina. Most insects, birds and reptiles have much better vision than humans. Bees see ultra-violet as well as blue and green light, so flowers would look quite different to them. Many birds have four types of cones; the fourth one picks up ultra-violet light. Pigeons have five types of cones and therefore are able to distinguish better between different 'colours'. Snakes see colour as well as infra-red, so they can perceive heat from other animals. Dogs have really poor colour vision. Cows see only black, white and shades of grey. Sharks have no colour vision either. Squirrels are red-green colour blind. Many simpler animals such as snails and jellyfish can only distinguish between light and dark; their 'pictures' of the surrounding are very crude.

Animal	Number of cone type	Black and white vision?	Colour vision?	Other vision?	Vision better or worse than human?

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Whose colour?

6 Write a report on light. Use the details below and other knowledge you have gained from this worksheet. First write a title. Then order the information into three paragraphs. In the first paragraph write about electromagnetic radiation in general. In the second paragraph, write about light. In the third paragraph, provide some historical information.

Let the scientist speak:

- Part of electromagnetic (EM) spectrum
- EM waves carry energy, travel at 300 000 000 m/s, can travel in vacuum, can travel through some media.
- Distinguished by wavelength, frequency, energy
- Main EM types in order of wavelength (long to short), frequency (low to high) and energy (low to high): radio waves, microwaves, infra-red waves, light, ultra-violet rays, X-rays, gamma rays.
- Light: Red, orange, yellow, green, blue, indigo, violet
- Wavelengths: red light: 700 nm (nanometre = 0.000 000 001 metre = 10^{-9} m); violet light: 400 nm
- Newton (1672) showed white light is a mixture of rainbow colours – a triangular prism splits white light into roygbiv; roygbiv colours combine into white light.

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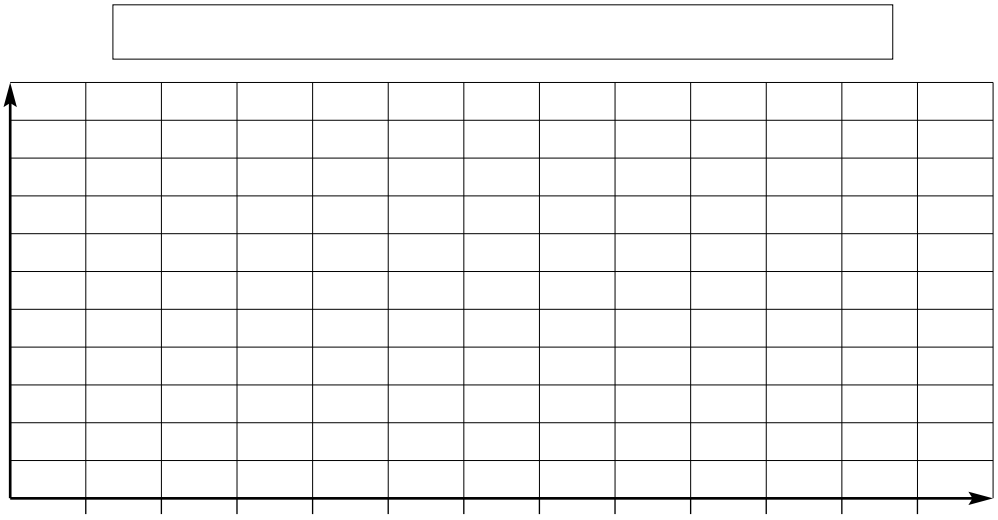
Hearing range and sound range

12a

Animals can hear sounds that are low pitch (i.e. low frequency) and also high pitch (high frequency). But the sounds they make often have a much smaller range.

Animal	Hearing range	Sound production range
Bat	2 to 110 kHz	
Beluga whale	100 Hz to 120 kHz	100 Hz to 120 kHz
Cat	70 Hz to 60 kHz	
Chicken	100 to 2 000 Hz	
Dog	50 to 50 000 Hz	
Elephant	1 to 12 000 Hz	5 to 9 000 Hz
Goldfish	20 to 3 000 Hz	
Human	20 Hz to 20 kHz	0.1 to 5 kHz
Mouse	1 to 100 kHz	
Moth	1 to 240 kHz	
Porpoise	1 to 150 kHz	1 to 150 kHz

1 Make a bar graph that shows the hearing range of the various animals. The horizontal axis represents the frequencies and the vertical axis lists the animals. Write a title



2 The low end of the frequency ranges cannot be read from the above graph. The graph on the next page uses an exponential scale for frequency. The first scale mark represents 10^0 (= 1), the second 10^1 (= 10), the third 10^2 (= 100) etc. Plot the hearing frequency range for the various animals using the exponential scale for frequencies. Add different coloured bars to indicate sound production ranges.

Key: blue = hearing range
red = sound production range

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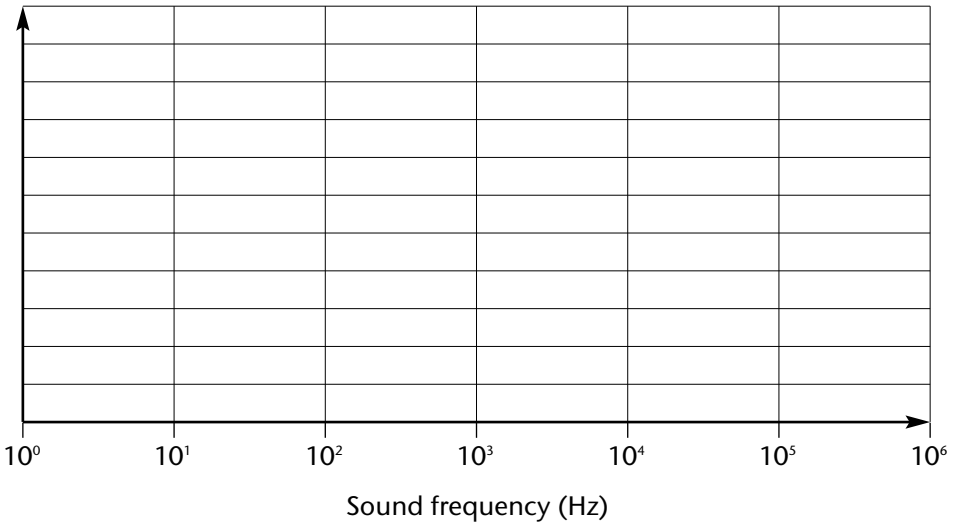
Hearing range and sound range

12b



3 Consider the animals listed on the previous page. Sort them by size.

Largest _____

_____ smallest

4 **a** *The bigger the skull, the lower the pitch an animal can hear.* Is this generalisation true? _____

b Make a more accurate generalisation relating animal size to hearing.

_____

_____

c Make another generalisation about the hearing range.

_____

_____

5 Infrasound is the name given to those sounds that are below the lowest limit of human hearing while ultrasound covers sounds that are above the human upper limit. Use the information in the above table and list the animals according to the following headings.

Hears infrasound and sound only	Hears sound and ultrasound only	Hears infrasound, sound and ultrasound

6 **a** A baby growing inside a uterus should not be 'looked at' with an X-ray machine. Explain.

_____

b Rather than use X-rays hospitals use ultrasound (frequency range of 1 to 20 MHz) to produce an image of babies. Rewrite 20 MHz in Hz. _____

c Rewrite 20 MHz in kHz. _____

d State which animals are capable of hearing the sound made by an ultrasound machine.

_____

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Seeing is believing

13a

Our eyes are one of our sense organs. They detect light.

On the outside, our eyes have protective **eyelids**. Closed, the eyelids protect eyes from smoke, onions, dust, bright lights and physical intrusion. The eyelids are surrounded by hairs called **eyelashes** and by touch receptors, that work together to protect our eyes.

Eyes are cleaned and lubricated by tears that are produced in the **tear gland**. Positioned near the eyebrows, the glands are not an external feature of the eye. Tears are released through **tear ducts** found near the corners of the eyes.

The coloured part of our eye is called the **iris**. It is a ring of muscle that changes the size of the **pupil**, the dark central area. Light passes through the pupil. If the light is too bright, the iris decreases the size of the pupil. If there is not much light, the iris increases the size of the pupil.

A cross-section of the human eye reveals a thin, clear skin, the **cornea**, that covers the front of the eye. Its properties allow light to pass through it. All the light that enters the eye is organised by the cornea and **lens**. The lens is made from a clear translucent material and is surrounded by a **lens muscle**. When the lens muscle relaxes, it becomes a large ring and the lens is made thin; when it contracts, it forms a small ring and the lens becomes thick. The cornea and lens focus the light rays onto the **retina**, which is located at the back of the eyeball. If a viewed object is close to the eye, the lens thickens so that the image of the object is focused and clear on the retina. For a distant object the eye lens is flattened. The retina has light receptors that are sensitive to the colours of light and to light brightness. The image of the object is upside down on the retina. The receptors in the retina send electrical impulses to the brain via the **optic nerve**, which is located outside the eyeball at the back of the eye. The brain interprets the impulses and 'turns the image the right way up'.

1 Complete the table. The place and function cells for the eyelid have been filled as an example.

Part	Place	Function
Eyelid	Outside, at the front	Protects the eye from things like dust, smoke, physical injury
Eyelashes		
Tear ducts		
Cornea		
Iris		
Pupil		
Lens		
Lens muscle		
Retina		
Optic nerve		

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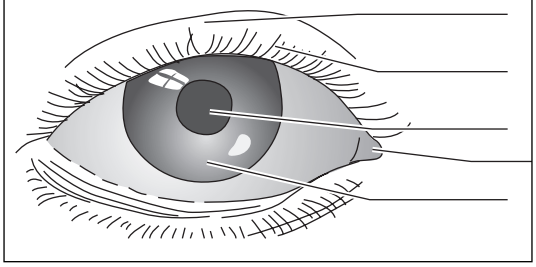
Seeing is believing

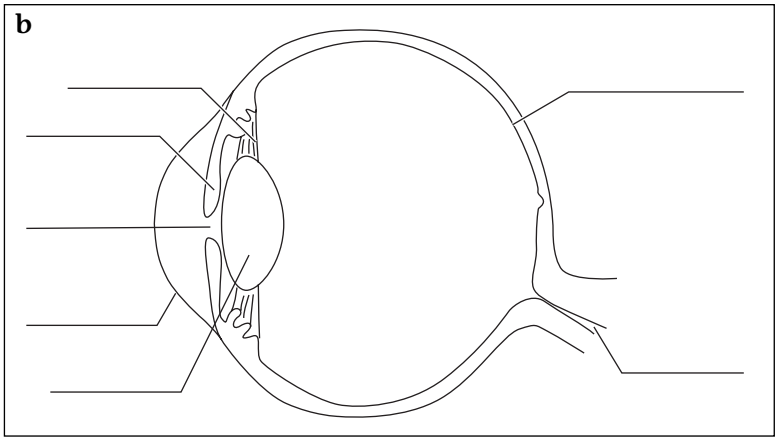
13b

2 Using words from the text above, label the arrowed features of each diagram.

a



b



3 Describe how the lens is adjusted when it focuses on an object that is:

a far away _____

b very close. _____

4 **a** What term could describe the type or style of text used at the beginning of this worksheet?

_____

b What is the purpose of this text type? _____

_____

c How many paragraphs does this text have? _____

d What does the first paragraph tell the audience? _____

e What does the rest of the text do? _____

_____

f A pronoun is a word used in place of a noun. We use pronouns (such as *they* and *it*) so we don't keep repeating the noun. Pronouns help to make the text clear to the reader. Underline three pronouns in the text about the eye, and draw an arrow to show which noun each underlined pronouns is replacing.

5 Use the text to help you unjumble the following sentences.

a an are detects Eyes important light. organ sense which

_____

b are by Eyes glands. in lubricated produced tears tear the

_____

c a clear cornea covers eye. front is of of part piece skin that the the The thin

_____

d image is on retina the The upside-down.

_____

e an and as eye focuses important incoming is it lens light. of organises part the the The

_____

_____

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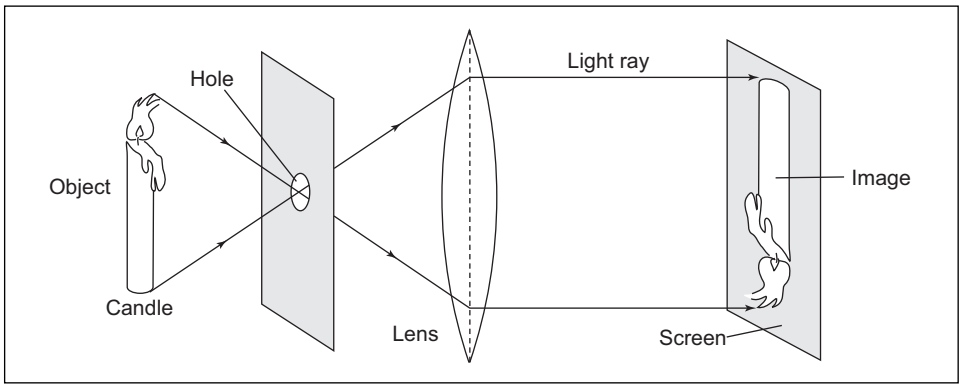
The Optometrist is in

14a

In your science lessons you may have used a candle flame to produce an inverted image when you were studying the eye.



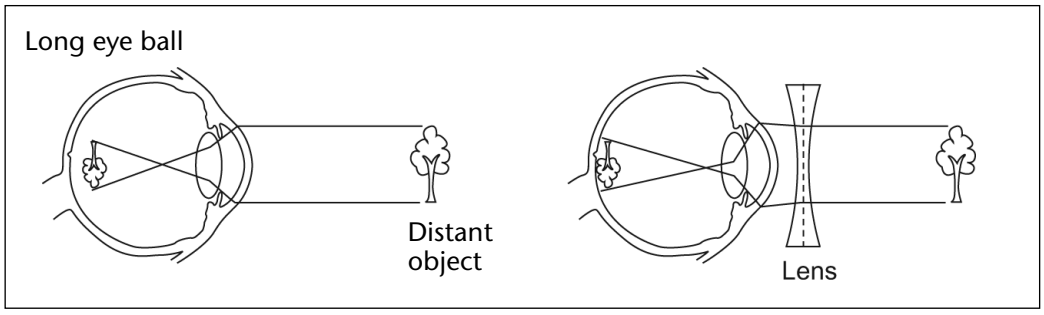
1 Explain what the diagram is representing. Use all the words in the diagram.

The lenses in your eyes perform a similar focus role. The image produced is focused at the back of your eye on the retina. Unfortunately some people have vision problems – a focused and clear image does not fall onto the retina. Long-sightedness and short-sightedness are two very common problems. Below is a table an optometrist developed to help explain to her patients the problems of long-sightedness and short-sightedness and how the problems could be corrected.

Disorder	Problem	Who can be affected	How is it corrected
Long-sightedness	Trouble focusing on close objects. Image focuses behind the retina. Can see distant objects clearly.	People with a short eyeball. People whose lens muscles have weakened or whose lenses have hardened.	Spectacles with converging lenses. These lenses cause light rays to bend inwards.
Short-sightedness	Hard to focus on objects far away. Image is focused in front of retina. Can see close objects clearly.	People who have a long eyeball. People whose lens muscles can no longer produce a thin lens.	Spectacles with diverging lenses. These lenses cause light rays to bend outwards.

2 Use information from the table to explain what is happening in each pair of diagrams.

a



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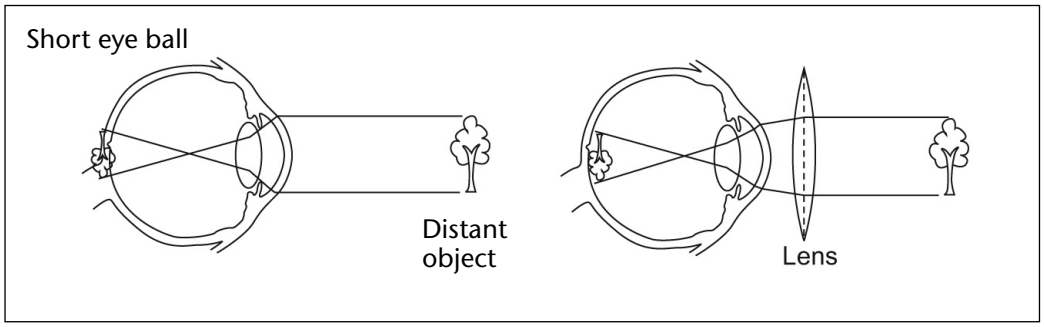
The Optometrist is in

b



3 Draw a diagram of a healthy eye (show the lens shape and where the image is projected) viewing an object that is:

a far away

b close to the eye.

4 As an optometrist, after examining one of your patients, you discover that she is suffering from long-sightedness. Use your knowledge and expertise in this area to explain to your patient why her vision is blurry and how you will correct it.

5 In your science book, rewrite the information from the table above as a report. Use the report text style. Remember to include a title and an introduction.

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Hearing

15a

Sound is produced by vibrating objects that cause the molecules in a transmitting medium (here taken as air) to vibrate. These vibrations are called sound waves. Sound waves travel through the air from a vibrating object to our ears.

Our ears can pick up different pitches. Pitches depend on the frequency of a sound wave – a high frequency vibration makes a high-pitched note, while a low frequency vibration makes a low-pitched note. Our ears can detect sounds ranging from 20 vibrations per second to 20 000 vibrations per second.

Below is a table detailing the different parts of a human ear.

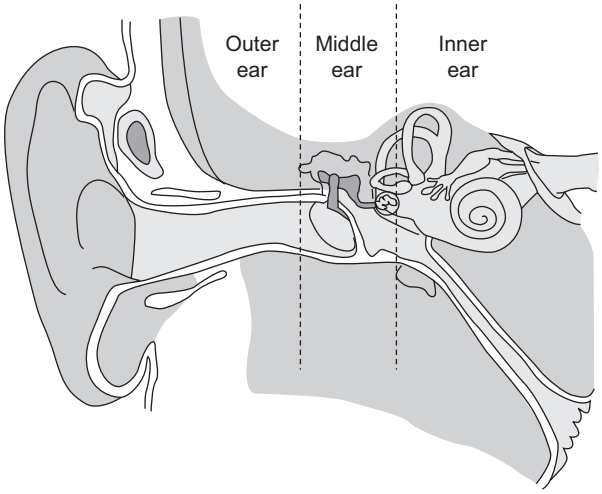
Part	Location	Function
Ear shell	Outer ear	Open funnel that collects sound waves
Ear canal	Outer ear	Long canal that leads sound waves to the ear drum
Eardrum	Middle ear	Thin membrane that vibrates when sound waves come in contact with it. It moves at the same frequency as the sound waves that hit it.
Hammer	Middle ear (surrounded by air)	Small bone near the eardrum. It picks up the vibrations from the eardrum and magnifies them.
Anvil	Middle ear (surrounded by air)	Hammer sends vibrations to the anvil, which is next to it. This is also a small bone that vibrates and magnifies vibrations.
Stirrup	Middle ear (surrounded by air)	Another small bone that picks up the vibrations from the anvil and magnifies them. It then vibrates against the inner ear.
Cochlea	Inner ear (filled with fluid)	Fluid filled tube that is coiled (looks like a snail shell). It contains receptors that detect vibrations in the fluid. Receptors send signals to the brain.
Semi-circular canals	Inner ear (filled with fluid)	Three tubes filled with fluid, each tube positioned in a different plane. They help with balance. Tiny hairs in the tubes trigger receptor cells that send signals about any movements to the brain.
Auditory nerve	Inner ear	Nerve that carries signals from the cochlea to the hearing centre of the brain. The brain then interprets these as sound.

1 **a** Use the table above. Which part of the ear has nothing to do with hearing? _____

b Explain why this part is very important. _____

_____

2 Use information from the table. Label the diagram of the ear.



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Hearing

3 In your science book, rewrite the information given in the table as a report. Don't forget a title and an introduction. Use the lines below to plan your report.

4 When you have a middle ear infection, the middle ear partly fills with a thick liquid. Explain why hearing would be difficult for you if this happened.

5 If there is an extrememly loud noise near you, your ear may not operate properly.

a Describe what could have happened to your eardrum and explain what exactly would have caused that.

b Explain why there could be a ringing in your ears that starts in the cochlea.

6 A tube connects your middle ear with your mouth cavity. When you fly up in a plane or drive up a long, steep hill, your ears 'pop' as you open your mouth or swallow. Assume that air moves only slowly in that tube and explain what causes your ears to pop.

7 Somebody hits you on the head and you hear a ringing or buzzing noise. Your ears themselves are in good working order and there is no such noise outside. Explain where the noise may come from.

8 When you spin on the spot for a couple of minutes, the liquid inside the semi-circular canals spins with you. But when you suddenly stop spinning, the liquid doesn't, and you feel as if you are spinning the opposite way. Explain why you feel dizzy and find it difficult to walk properly.

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Seeing and hearing

16a

1 Our eyes and ears are sense organs. Name three other sense organs.

_____

2 Sense organs respond to a stimulus. Eyes and ears respond to different waves. The eye responds to _____ and the ear responds to _____.

3 **a** State which type of wave always requires a medium for carrying it.

_____

b State the type of wave that does not always require a transmission medium.

_____

4 Match each ear part with a description and a function, by writing the letters identifying the Description and Function in the Label box. (Add letters to the label cells.)

Labels

1 Anvil __ __
2 Auditory nerve __ __
3 Cochlea __ __
4 Ear canal __ __
5 Ear shell __ __
6 Eardrum __ __
7 Eustachian tube __ __
8 Hammer __ __
9 Stirrup __ __
10 Window __ __

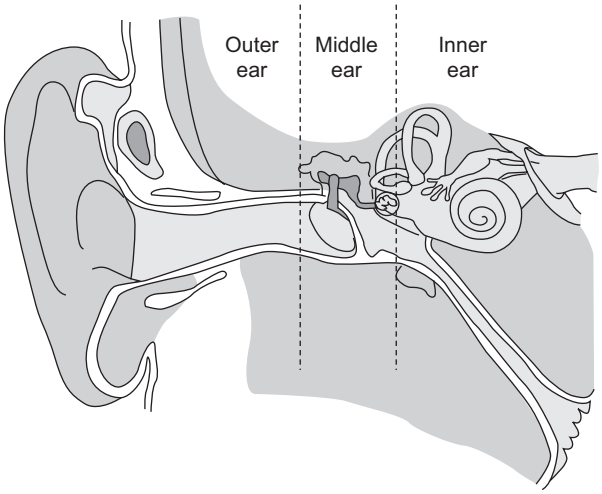
Descriptions

A Shell-like cartilage structure
B Small thin membrane at beginning of cochlea
C Snail-like structure containing a liquid
D Thick nerve linking cochlea and brain
E Thin membrane at end of ear canal
F Thin tube between middle ear and mouth
G Tiny ear bone attached to eardrum
H Tiny ear bone attached to window
I Tiny ear bone between hammer and stirrup
J Tube between ear shell and ear drum

Functions

a Changes vibrations in stirrup to vibrations inside cochlea
b Channels sound waves to eardrum
c Collects sound, directs it to ear canal
d Equalises air pressure between middle ear and mouth
e Holds nerve endings which are sensitive to vibrations
f Magnifies sound vibrations
g Magnifies sound vibrations
h Magnifies sound vibrations
i Sends electrical pulses from nerve endings in cochlea to brain
j Vibrates when sound waves hit it, makes hammer vibrate

5 Add labels to the diagram of the human ear. (All the names can be found in the boxes above.)



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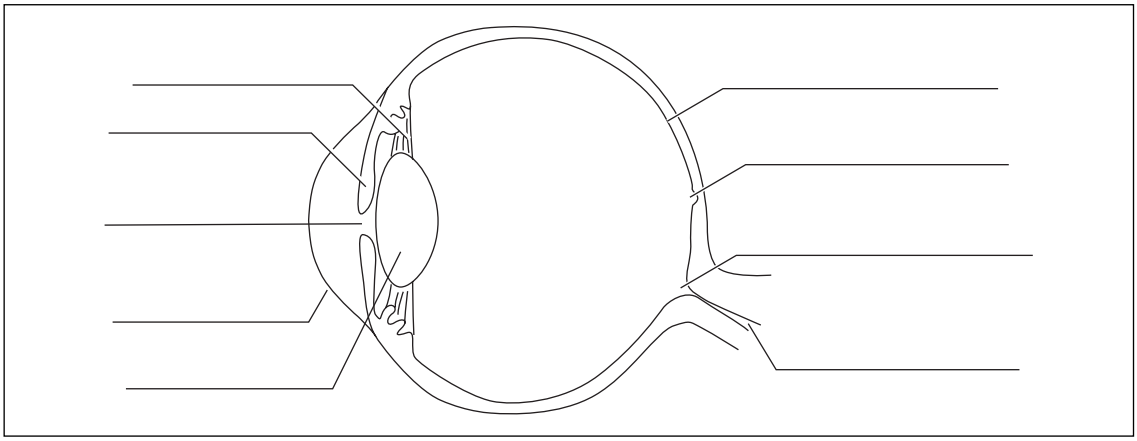
Seeing and hearing

6 Match an eye part with a description and a function. (Add letters to the label cells.)

Labels		Descriptions	Functions
1 Blind spot ___ ___		**A** Area where most light rays hit retina	**a** Adjusts lens for better focusing
2 Cornea ___ ___		**B** Area where optic nerve leaves eye	**b** Allows light to enter eye
3 Fovea ___ ___		**C** Clear skin at front of eye	**c** Carries electrical impulses from retina to brain
4 Iris ___ ___		**D** Clear soft thick disk	**d** Controls amount of light entering eye
5 Lens ___ ___		**E** Colourful broad ring in front of lens	**e** Changes light into electrical impulses
6 Lens muscle ___ ___		**F** Opening controlled by iris	**f** Does not have light sensitive nerve endings, does not 'see'
7 Optic nerve ___ ___		**G** Light sensitive layer inside eye	**g** Has many light sensitive nerve endings and 'sees' well
8 Pupil ___ ___		**H** Ring muscle around lens	**h** Helps focus light rays onto retina
9 Retina ___ ___		**I** Thick nerve linking eye and brain	**i** Protects eye, helps focus light rays onto retina

7 Label the diagram of a human eye. (The names of the parts are in the table above.)



8 State in which way the following are similar:

a Optic nerve and auditory nerve _____

b Retina and cochlea _____

c Lens and hammer, anvil, stirrup. _____

9 Contrast the following:

a Retina and cochlea _____

_____

b Lens and hammer, anvil, stirrup _____

_____

10 Parts in our eye allow us to control things. In your science book, list several parts of the eye you labelled in question 7, and state what they control.

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A history of the telescope

17a

3 500 BC When Phoenicians were cooking on sand around 3 500 BC, they discovered glass. But it took thousands of years before glass was shaped into a lens for the first telescope. Lenses that could change the magnification of objects were known in Europe at the end of the thirteenth century.

1608 The spectacle maker Hans Lippershey (1570–1619), born in Germany and working in Holland, is often credited with the invention of the first telescope in 1608. If one placed a biconvex lens (a lens used to correct long-sightedness) about 40 cm away from a biconcave lens (a lens used to correct short-sightedness) and then looked through them, distant objects appeared closer. This spyglass was used for spotting the army's enemy.

1609 The telescope was introduced to astronomy in 1609 by the Italian scientist Galileo Galilei (1564–1642), who became the first man to see the craters of the moon and went on to discover sunspots, the four large moons of Jupiter and the rings of Saturn. Galileo's telescope was a simple refractor telescope, using a plano-convex lens and a plano-concave eye lens in a tube. This arrangement magnified up to 30 times but provided only a narrow field of view: Galileo could see no more than a quarter of the moon's face. The German Johannes Kepler (1571–1630) invented the form of the refracting telescope that is the basis for modern refractors; it has a convex eye lens. However, Kepler's telescope produced large inverted images with some chromatic distortions (the lenses acted like a prism and split the starlight into its constituent colours resulting in coloured fuzzy edges of the images).

1704 In 1704, Sir Isaac Newton announced a new concept in telescope design: instead of lenses, a curved mirror was used to gather in light and reflect it. This reflecting mirror acted like a light-collecting bucket. Newton's reflector telescope could magnify objects much more than could ever be obtained with a lens and without chromatic distortion. The reflecting telescope invented by Isaac Newton used a large spherical mirror at the end of a tube to reflect the image towards the open end where a small mirror tilted at 45° then projected the image to a convex eyepiece lens sitting on the side of the tube.

1974 Telescopes grew from the 15 cm mirror used by Newton to the 6 m mirror of the Special Astrophysical Observatory in Russia, which opened in 1974.

1930 Bernhard Schmidt (1879–1935) was born in Estonia, studied in Germany and worked as an optical technician at the Bergedorf Observatory, Hamburg. In 1930 he invented the telescope that is named after him. It uses a spherical mirror, not a parabolical reflector, and a correcting lens at the telescope opening to compensate for spherical aberration; thus it is a combination reflector-refractor system. The Schmidt telescope is the most popular among amateur astronomers because of its compact design and large aperture, and because the optics are completely enclosed. The largest Schmidt telescope is located at Palomar Observatory (California). It can photograph an area of the sky more than 300 times as large as that seen by other reflectors.

1977 Unfortunately the atmospheric layers have hampered telescope observations from the Earth's surface. Scientists, therefore, have developed other more efficient means for gathering data about the heavens. Some of the largest reflective telescopes are located high up on Palomar Mountain near San Diego, California, and on Mount Pastukhov in Russia. Scientists have carried telescopes in planes: the Kuiper Airborne Observatory, a cargo plane fitted with a 200 kg infrared telescope, is operated at about 15 km altitude. The Kuiper discovered the rings of Uranus in 1977 and the atmosphere around Pluto in 1988.

1990 Since 1990 the Hubble Space Telescope, run jointly by NASA and the European Space Agency, is orbiting the Earth at an altitude of 600 km and capturing images from what scientists believe is the edge of our universe.

1937 Radio waves are also a part of the electromagnetic spectrum. Just as an optical telescope gathers visible light and magnifies it, a radio telescope is a large dish-shaped device that gathers and amplifies radio frequencies from space. Radio telescopes do not 'see' the way optical telescopes do; the radio signals that the telescopes gather are used to create diagrams showing where the strongest radio sources are located. Grote Reber (1911–2002), an American amateur ham-radio enthusiast, built the world's first radio telescope in his backyard in Chicago, Illinois, in 1937. Reber's device was similar in design to the backyard satellite dishes of today, though of a considerably larger scale. If the dish is bigger, more data can be collected, and the resolution and sensitivity of the telescope is greater.

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A history of the telescope

17b

1947 In 1947 Bernard Lovell (1913–) directed the construction of a parabolic radio reflector 66 m in diameter at Jodrell Bank, England. It was also there that a 76 m dish went into operation in August 1957.

1962 The 91m–diameter dish at Green Bank, West Virginia, built by the National Radio Astronomy Observatory in 1962, was the largest steerable radio telescope in the world, until it collapsed from metal fatigue in 1989.

1963 The largest individual dish currently in existence is in Arecibo, Puerto Rico. This radio telescope was created out of a natural depression within a ring of mountains in 1963. With a diameter of 305 m, the telescope's reflecting surface is too large to be steerable. Instead, a movable feed suspended above the dish provides access to most of the sky.

Perform the tasks below and answer the following questions in your science book.

Sketches

1 Sketch the following lenses: plano-convex, plano-concave, biconvex and biconcave.

2 Sketch the following mirrors: plane, spherical and paraboloid.

3 Sketch the telescopes built by Lippershey, Galilei, Kepler, Newton and Schmidt. Pay particular attention to the shape of the lenses and mirrors, as much as you can.

4 Sketch Reber's radio telescope.

On the lines

5 When did lenses appear first?

6 When and where was the first telescope built?

7 About how long was Lippershey's telescope?

8 What was the first telescope used for?

9 What is a refractor telescope made of?

10 How many times did Galilei's best telescope magnify heavenly objects?

11 How did Newton's telescope present a new concept?

12 What is chromatic distortion?

13 Why was Newton's telescope superior to Kepler's?

14 What was the advantage of Schmidt's telescope over Newton's?

Between the lines

15 How long ago was glass discovered?

16 Where did Lippershey get his lenses from for the first telescope?

17 As Galilei's telescope magnified more and more, it had to become longer and longer. What restriction resulted from having better magnification?

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A history of the telescope

18 How is Kepler's telescope different to Galilei's?

19 How did Newton overcome the problem of chromatic distortion?

20 How many years passed between the invention of the refractor telescope and the invention of the reflector telescope?

21 How many times is the mirror of the Special Astrophysical Observatory in Russia bigger than the one in Newton's telescope?

22 How is Schmidt's telescope a reflector-refractor telescope?

23 Why are today's telescopes mounted at high altitudes?

24 If larger telescopes produce better images, why don't scientists build them in kilometre sizes?

Beyond the lines

25 Where did the Phoenicians live?

26 If Galilei was the first person to see the craters on the moon, what did people before him see on the moon?

27 Galilei found that Jupiter had moons orbiting it. Why would this have caused interest among scientists and people? (Hint: People then believed in a geocentric universe.)

28 Newton's telescope had a large curved mirror and a small tilted one. Why was this second mirror small and tilted?

29 What in the atmospheric layers might hamper astronomical observations?

30 Why were Galilei, Kepler and Newton not concerned about the interference of the atmosphere during their telescope observations?

31 The Hubble Space Telescope sent pictures to Earth that were slightly out of focus, because the mirror was too flat at one edge by 1/50 of the width of a human hair. Why didn't the scientist notice this fault before launching the telescope into space?

32 How would the Hubble Space Telescope be repaired, if there were problems with its operation?

33 What else is part of the electromagnetic spectrum apart from light and radio waves?

34 How are the different wave types of the electromagnetic spectrum different and how are they similar?

Developing a timeline

25 On a separate sheet of paper, draw a timeline that can show all the dates in the three pages of this worksheet. Label it using thousands of years. Then add to it the names and short terms representing what you think are the major events in the history of the telescope.

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The facts of light – A game

18a

Getting started

- Photocopy the gameboard, trim the inside edges and tape it together.
- This game is for two to four players.
- Each group of players needs one gameboard and one dice.
- Each player needs one token, a pen and a copy of the table below.

The rules

- The aim of the game is to complete the table below.
- At the beginning of the game all players place their tokens at the START cell on the board.
- Each player throws the dice and the player with the highest number has first move.
- Players in turn throw the dice and move that number of spaces.
- You can move in any direction, but you cannot land your token on a cell that is occupied by the token of another player.
- When you land on a cell, you may enter the text from that cell in the appropriate column of the table below.
- The challenge is to fill each of the table cells. To do this, you need to select five definitions/terms related to light, five properties and five uses of light. The definitions, properties and uses do not have to match.
- The first player to complete the table wins!

Table

Definitions/related terms	Properties of light	Uses of light

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The facts of light – Gameboard

18b

	Light helps us to detect what goes on around us. Green plants also use light to make their own food.		Bending of light rays with lenses has enabled us to build useful optical instruments like cameras, telescopes, microscopes.	
Concave lenses form an image that is magnified and upright if viewed close.		Ask another player to draw a concave lens. If incorrect, that player misses a turn.		Irregular reflection occurs when rays of light strike rough surfaces and reflected rays are scattered.
Lenses bend light in organised ways.		Normal	Absorbed	A rainbow shows that sunlight is made up of seven colours.
	A shadow is an area of darkness or partial darkness, formed when light paths are blocked.		You broke a convex lens. Draw it correctly or miss a turn.	Opaque
When light passes from one medium into another, its speed changes causing the light to bend or refract.	White light can be separated into the rainbow colours using a triangular prism.	Luminous		Eye lenses change shape to focus images on the retina.
In cameras lenses are used to form clear images on a light sensitive surface.	Incident rays	Long-sightedness		Microscopes are important – increasing knowledge about ourselves and our surroundings.
Convex lenses form an image that is upright and small.		Light travels at a speed of 300 000 000 m/s.	The law of reflection states that the angle of incidence equals the angle of reflection.	

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The facts of light – Gameboard

18c

	Light is electromagnetic energy that enables us to see.		Lenses in front of our eyes help overcome eye problems like short-sightedness and long-sightedness.	
Visible spectrum		Light waves are electromagnetic waves with wavelengths between 3.5×10^7 and 7.5×10^7 m.		The primary colours are red, green and blue. Other colours can be made by combining these.
Object colours depend on the parts of white light that the object reflects.		You have learnt a lot about light. Have another turn.		White light is a mixture of colours.
	Retina			
START			When light lands on the surface of a material, it might be reflected, passed through or absorbed.	
Reflected		Transparent	Lenses are used in projectors to throw an image on a screen.	
	The sun produces its own light. It is a luminous object.		Traditional TV sets use focused beams of electrons. Magnets are used to focus them rather than a glass lens.	Optical
Lenses are transparent and have one or both sides curved.		Optic fibres are long thin glass cored tubes. Light is internally reflected in the tube. Fibre optics is important in information transfer.		Translucent

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Answers

1: IMAGES IN A CONCAVE AND IN A CONVEX MIRROR

1 **a** The image is closer, smaller, inverted and real. **b** The image is further, larger, inverted, and real. **c** There is no image. **d** The image is further, bigger, upright, virtual.

2: OBJECTS AND THEIR IMAGES IN A LENS

1 **a** The image is closer, smaller, inverted and real. **b** The image is further, larger, inverted, real. **c** There is no image. **d** The image is further, bigger, upright, virtual.

3: THE ELECTROMAGNETIC SPECTRUM

1 Radio, micro, infra-red, light, ultra-violet, X, gamma.
3 Radio, micro, infrared, light, ultra-violet, X, gamma.
4 Identical lists.
5 Radio, micro, infrared, light, ultra-violet, X, gamma.
6 Identical lists
7 House, bee, pinhead, plant cell, molecule, atom, nucleus.
8 Radio transmitter, microwave oven, electric radiator, light bulb, quartz lamp, X-ray machine, reactor.
9 Communication, cooking, heat, sight, skin cancer, deformities, death.
11 atmosphere, layer, absorb, ozone, during, decades, develops, one, rebuild, Unfortunately, thin, Australia, early, Sunscreen, clothing, protection, dangerous, believed, loss, increase, depletion, hole, increase, populations, Asia, would, by, ultraviolet.

4: TYPES OF WAVES

1 Transverse, longitudinal.
2 Longitudinal, as particles (i.e. molecules) bump against each other and so send the sound further along.
3 Iron, because the particles are very close together and already vibrating.
4 The speed drops. Air molecules are further apart so the bumping to pass on the sound takes place more slowly. Or As there are fewer air molecules they are less likely to bump into each other.
5 No air particles at all.
7 **b** 28 mm
8 **b** 9 mm
9 Middle C: 256/1 = 256 Hz; A: 880/2 = 440 Hz; D: 864/3 = 288 Hz; High C: 2048/4 = 512 Hz; E:1400/5 = 320 Hz; F: 341/1 = 341 Hz; G: 678/2 = 384 Hz; A: 660/3 = 220 Hz; B: 960/4 = 240 Hz; A: 550/5 = 110 Hz; B: 960/2 = 480 Hz
10 A (220 Hz), B, C, D, E, F, G, A
11 440 Hz, 220 Hz, 110 Hz
12 Frequency doubles for the higher A.
13 Medium: empty space, air, water, glass, diamond
Speed of light: 300 000 000 m/s, 300 000 000 m/s, 225 000 000 m/s, 200 000 000 m/s, 125 000 000 m/s.
14 Air molecules move more slowly at lower temperatures, therefore they bump into each other less frequently.
15 Light travels more slowly in denser materials.
16 Sound travels faster in denser materials.

5: SPEED OF LIGHT AND SOUND

1 Particles in an iron bar are much closer together therefore bumping into each other faster or more often.
2 Hot air particles move faster and so bump into each other more often.
3 Molecules are closer together in glass, therefore more molecules need to absorb and release light energy.
4 Very few molecules absorb and release light energy, thus hardly slowing down the light waves.
5 **a** Light: 300 000 000 m/s = 300 000 000 . 1/1000 = 3000 000 000/1000 = 300 000 km/s
Sound: 333 . 1/1000 = 333/1000 = 0.3 km/s
b Light: 300 000 km/s = 300 000 . 3600 = 1 080 000 000 km/h
Sound: 0.333 km/s = 0.333 . 3600 = 1 199 km/h
6 It takes practically no time for the light to travel from smoke to timekeeper. However, the sound of the gun takes a little time (a fraction of a second).
7 By the time the plane sound is heard, the plane has flown further forward. The sound takes a little time to travel from plane to listener, while light travels from plane to observer in practically no time.
8 speed = distance/time and therefore time (s) = distance (m)/speed (m/s) = 6000/333 = 18 s
9 speed = distance/time and therefore distance (m) = speed (m/s) . time(s) = 333 . 3 = 999 m (or 1 km)
10 speed = distance / time therefore time = distance / speed = 3 000 000 000 m / 300 000 000 m/s = 10 s You would see the explosion 10 s after the event. You would never hear the explosion because sound does not travel through (empty) space (no molecules to vibrate!).
11 m/s, m, Hz
12 **a** speed = wavelength . frequency
b wavelength = speed/frequency **c** frequency = speed/wavelength.
13 Middle C: wavelength = speed/frequency = 342/256 = 1.3 m; High C: 342 / 512 = 0.7 m; Low C: 342 / 128 = 2.7 m
Concert pitch: 342 / 440 = 0.8 m; Lowest: 342 / 20 = 17.1 m; Highest: 342 / 20 000 = 0.02 m.
14 300 000 000 m/s, 300 000 km/s

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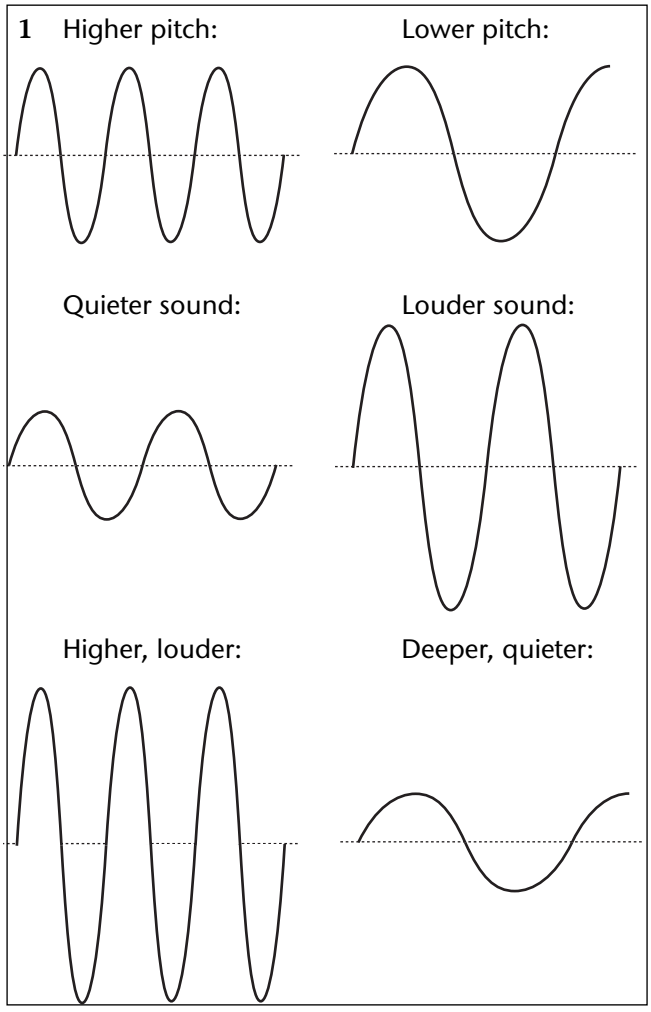
Answers

15 Gamma: wavelength = speed/frequency
$= 3 . 10^8/3 . 10^{19} = 3/3 . 10^8/10^{19} = 1 . 10^{8-19}$
$= 10^{-11}$ m
Red: frequency = speed/wavelength =
$3 . 10^8 /750 . 10^{-9} = 3/750 . 10^{8+9} = 4 . 10^{-3} . 10^{17}$
$= 4 . 10^{17-3} = 4 . 10^{14}$ Hz
Blue: wavelength = speed/frequency
$= 3 . 10^8/6.7 . 10^{14} = 3/6.7 . 10^{8-14} = 4 . 10^{-1} . 10^{-6}$
$= 4 . 10^{-7}$ m
Radio: frequency = speed/wavelength
$= 3 . 10^8/10^2 = 3 . 10^{8-2} = 3 . 10^6$ Hz

7: SPEAKING AND LISTENING WITH A TELEPHONE



2 Soft and deep: lower, lower, weaker, lower
Soft and high: higher, lower, weaker, higher
Loud and high: higher, higher, stronger, higher
Loud and deep: lower, higher, stronger, higher

3 From left, top, clockwise: free-moving plate, microphone box, fixed plate, wires, carbon powder.

4 From the top, right, clockwise: box, cardboard, iron plate, electromagnet, wires.

8: SENDING AND RECEIVING MUSIC BY RADIO

1 The matchings are: (1, B), (2, E), (3, A), (4, D), (5, C), (6, F) and (7, G).

3,5 From the top left, clockwise: C - antenna, A - tuner, D - speaker, E - amplifier, B - demodulator.

4 The order is: (1, C), (2, A), (3, B), (4, E) and (5, D).

6 C: The antenna picks up radio waves; it generates a weak alternating current.
A: The tuner selects an alternating current of a particular frequency and passes it on.
B: The demodulator separates the carrier wave from the signal it carried and passes only the audio wave on.
E: The amplifier amplifies the audio wave to a desired loudness.
D: The speaker converts the amplified audio wave into sound and music.

9: REFLECTION AND REFRACTION - A CONTRAST

1 Such as: although, however, but, and not, not the same as.

2 Such as: A concave mirror curves in and away from you, while a convex mirror curves in and towards you.

3 Such as: A concave lens is thin in the middle, however a convex lens is thick in the middle.

4 **a** Incoming light wave **b** Light wave that bounced back after hitting the mirror **c** Light wave that has come through a lens and been bent **d** An imaginary line sticking out at a right angle from the mirror or lens where the light ray hits it. **e** Angle between the incoming light ray and the normal **f** Angle between the reflected light ray and the normal **g** Angle between the refracted light ray and the normal

10: REFLECTION, REFRACTION, DIFFRACTION, SCATTERING

1 **a** Transmission, light **b** Transmission, light or sound.
2 Refraction or transmission, light.
3 Scattering, light
4 Reflection, light.
5 Reflection, light.
6 Reflection, sound.
7 Reflection, light.
8 Reflection, light.
9 Refraction, light.
10 Refraction, light.
11 Refraction, light.
12 Scattering, light.
13 Refraction, light.
14 Reflection, sound.
15 Reflection, light.
16 Refraction, sound.
17 Refraction, light.
18 Diffraction, sound
19 Diffraction, water.
20 Reflection, light.

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Answers

11: WHOSE COLOUR?

1 Correct words are: colours, create, effects, lights, shining, spot, blue, produced, yellow, three, together.

2 The tenses are: are made up of, are stimulated, glow, are used, can be produced, stimulate, have, are stimulated, are, merge, are lit up, result, are, would appear, might be able to see, look, has, resulting.

3 The prepositions are: from, on, from, with, in, in.

4 Paragraph 1: (1,1a), (2, 1b). Paragraph 2: (1, 2d), (2, 2a), (3, 2c), (4, 2b). Paragraph 3: (1, 3a), (2, 3e), (3, 3f), (4, 3d), (5, 3b), (6, 3c).

12: HEARING RANGE AND SOUND RANGE

3 Beluga whale, elephant, porpoise, human, dog, cat, chicken, bat, goldfish, mouse, moth.

4 **a** No **b** e.g. Animals with big and small skulls can hear very low pitch sounds. **c** e.g. Very big and very small animals have wide ranges of hearing.

5 Infrasound: Elephant; Ultrasound: Whale, dog, cat; All: Porpoise, bat, mouse, moth.

6 **a** X-rays may damage the baby by interfering with cell division. **b** 20 MHz = 20 000 000 Hz. **c** 20 MHz = 20 000 kHz. **d** Elephant, porpoise, bat, mouse, moth.

13: SEEING IS BELIEVING

1 **Eyelashes:** outside, in front, on eyelids – help to protect eyeballs.
Tear ducts: outside, near the corners – transport tears.
Cornea: at the front – organises light travelling through.
Iris: near the front – changes the size of the pupil
Pupil: near the front, inside iris – controls amount of light passing through
Lens: in the middle – focuses light onto retina
Lens muscle: in the middle, around the lens – makes lens thick or thin
Retina: at the back – changes light into electrical impulses
Optic nerve: outside, at the back – sends electrical impulses to brain

2 **a** From top: eyelid, eyelashes, pupil, tear duct, iris.
b Clockwise from top right: retina, optic nerve, lens, cornea, pupil, iris, lens muscle

3 **a** Lens muscle relaxes causing the lens to become thin
b Lens muscle contracts making the lens thick

4 **a** Report **b** Describes the eye and its parts. **c** Five **d** That the text is about the eye, what group the eye belongs to **e** Describes the various parts of the eye: where they are, what they look like, what they do.

5 **a** Eyes are an important sense organ which detect light. **b** Eyes are lubricated by tears produced in the tear glands. **c** The cornea is a clear thin piece of skin that covers the front part of the eye. **d** The image on the retina is upside-down. **e** The lens is an important part of the eye as it organises and focuses the incoming light.

15: HEARING

1 **a** Semi-circular canals **b** Tells head position and controls balance

4 Hammer, anvil and stirrup can't move freely and impair the transfer of sound.

5 **a** Your eardrum may be ruptures because of the very strong/ intense sound wave.
b Over-stimulated nerve endings in the cochlea keep sending off 'noise' signals to the brain.

6 The air pressure outside is different to the air pressure in your mouth and middle ear, as a result the eardrum bulges out or in. Swallowing equalises the pressure suddenly and the eardrum snaps back into normal position.

7 The brain is not working properly.

8 The liquid keeps spinning although you are standing still. Therefore the semi-circular canals send a message to the brain saying that your body is spinning and as a result your body is confused.

16: SEEING AND HEARING

1 Nose, taste buds and skin

2 Light waves, sound waves

3 **a** Sound OR longitudinal OR compression, light OR electromagnetic

4 1 (I, f), 2 (D, i), 3 (C, e), 4 (J, b), 5 (A, c), 6 (E, j), 7 (F, d), 8 (G, g), 9 (H, h), 10 (B, a)

6 1 (B, f), 2 (C, i), 3 (A, g), 4 (E, d), 5 (D, h), 6 (H, a), 7 (I, c), 8 (F, b), 9 (G, e)

8 **a** Carry electric impulses away from a sense organ to brain **b** Convert stimulus (i.e. light, sound waves) into electrical impulses **c** Make stimulus better for sense organ

9 **a** Retina converts light waves into electrical impulses, while cochlea converts sound waves.
b Lens focuses light rays, ear bones magnify sound waves.

17: A HISTORY OF THE TELESCOPE

1

2

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